

Nano additive enhanced salt hydrate phase change materials for thermal energy storage

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ABSTRACT

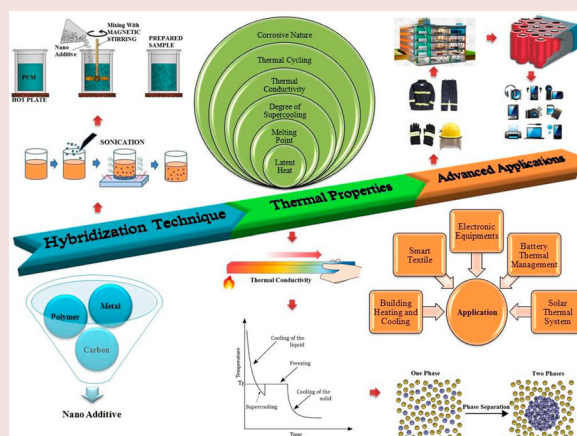
Energy storage plays a vital role in sustainable development. Focus on energy storage using phase change materials (PCMs) are of current research hotspot due to high latent heat value. Nevertheless, poor thermal conductivity, supercooling, phase separation, corrosive nature of salt hydrate is of great concern. Distress related to properties of PCM is resolved using nano additives. Major research focus on the dispersion of nano additive with PCM depends on (a) technique of preparing a novel composite PCM; (b) improvement of their thermophysical characteristic; and (c) advanced application for human comfort without polluting the environment. This article presents a critical review of hybridization techniques of metal, carbon and polymer additives on salt hydrate PCMs. To facilitate researchers, the significant variation on thermophysical properties of salt hydrate with nano additives are vitally compared, analysed and critically reviewed. This review article also includes the advanced application of nano additives-based salt hydrate PCMs.

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KEYWORDS

Salt hydrate PCMs; hybridization; nano additives; carbon nanoparticle; thermal property; thermal energy storage; PCM applications; inorganic phase change material



Highlights

- Salt hydrates are cheaper than organic PCM and exhibit good thermal properties for TES.
- Synthesis and hybridization techniques of composite salt hydrate are elaborated.
- Energy storage potential increases with the dispersion of nano additives.
- Nano additive effects on thermal and chemical stability are emphasized.
- Applications of nano additive PCM with 0–350°C PTT are presented.

Introduction

Clean, affordable, reliable and enduring supply of energy is the basic need for all developing countries [1]. Rapid growth in a global economy and industrial development highly depends on energy consumption. High consumption of fossil fuel has elevated the alert of depletion. As well, electric power generation using fossil fuel causes environmental pollution by emitting greenhouse gases. These harmful emissions raise the ambient temperature of the environment and cause concern over the climatic condition. In the year 2015, United Nations member states introduced 17

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sustainable development goals (SDG) to be achieved in 2030 [2]. Among the 17 sustainable goals, this article mainly contributes towards the seventh SDG in terms of techniques that help to ensure dependable and sustainable energy for all.

Thermal energy storage and its significance

Thermal energy storage (TES) is a technique to store heat or cold energy within materials for later use under different temperatures [3]. TES system consists of heat storage medium (stores energy in the form of latent or sensible heat), energy transfer method (liberate and absorb the required amount of heat) and a containment system (to provide proper insulation and leak proof) [4]. Sensible heat depicts the amount of heat that is added or removed from a substance during temperature change. Latent heat describes the amount of heat that is added or removed from a substance during phase change at a constant temperature. The storage is a near-isothermal process that can provide significantly high storage density with smaller temperature swings in comparison with sensible storage systems. In addition, latent heat storage has the capacity to store the heat of fusion at a constant or near-constant temperature that corresponds to the phase transition temperature (PTT) of the phase change material (PCM). TES enhances solar power utilization through the storage facility. It also bridges the gap between energy supply and demand by prevailing over the intermittent nature of solar power [5]. When the temperature of materials like water, air, concrete, ceramic, sand, brick, bedrock and cast iron are increased or decreased by external heat source sensible heat storage occurs [6]. Latent heat storage undergoes phase transition of material from solid–liquid on melting and freezing, liquid–gas on evaporation and condensation, solid to gas on sublimation and deposition or solid–solid with change in

crystalline structure [3,7] vice versa. Phase changes, such as vaporization and melting, are accompanied by relatively large changes in internal energy and enthalpy as bonds between molecules are broken and reformed [8]. Heat absorbed during a phase change is used to modify the microscopic structure of the substance. Heat absorbed by a solid during melting is used to break the bonds between atoms and a tiny fraction of heat absorbed is converted into work (PdV – where P is the pressure and dV is the change in volume) which is associated with volume change [9]. Thus the potential of latent heat is highly linked with the number of breakage in chemical bonds. The phase transition from solid to gas and from liquid to gas phase offer higher latent heat as they undergo dominant volume change during phase transition, as well they require closed storage containers [10]. PCM has well-developed technology towards heat liberation and absorption during isothermal phase change and can be actively used for TES [11,12]. They are also on the verge of versatile development over the past decade. The most commonly used PCM falls under the category of solid–liquid phase transition. TES using PCM provides numerous notable significances, as shown in Figure 1. Thermal energy is used widely in domestic and industrial applications for heating. Integration of PCM for TES is an interesting phenomenon, where PCM are filled within a storage system. High-temperature fluids are passed through the pipeline on the storage system. PCM absorbs heat energy from the high-temperature fluids until it completely transfers its phase from solid to liquid. During thermal energy need, low-temperature fluids are passed through the storage system, the PCM liberate heat and raises the fluid temperature. TES using PCM is accompanied with a storage system for its integration.

Role and dominant properties of PCM

PCM with large latent heat values during phase transition are commonly preferred for TES. With the rise in surrounding temperature of PCM's, the PCM slowly reaches its melting point, at this junction, the chemical bonds start to break with the endothermic process and absorb heat energy with a change in state from solid to liquid. This process is called PCM charging. Similarly, with a decrease in the surrounding temperature of PCM's, the PCM reaches its freezing point, at this junction, the chemical bonds regenerate with the exothermic process and liberate heat energy with change in state from liquid to solid. This process is called as PCM discharging. The effectiveness of TES using inorganic PCM highly depends on thermophysical, chemical and economical properties. Inorganic PCMs are classified as salt hydrates and metals & metallic alloys. Salt hydrate PCM faces a major issue of

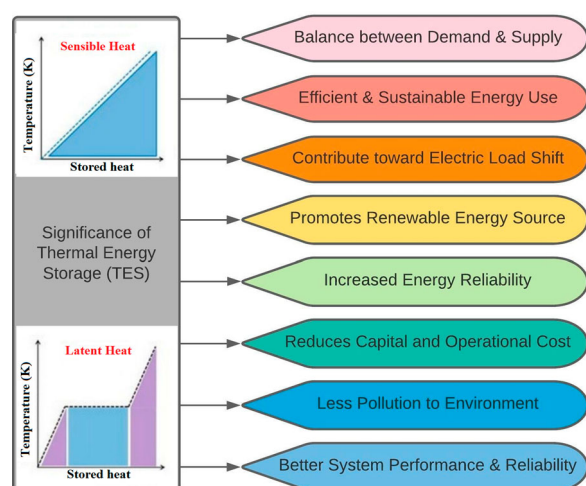


Figure 1. Notable significance of thermal energy storage.

supercooling, phase separation, lack of resistance to corrosion and low thermal conductivity. The phenomenon during which the salt hydrate PCM exists in a liquid state even after reaching their fusion temperature or freezing temperature is known as supercooling or degree of supercooling, commonly supercooling is represented in terms of temperature. Thermal conductivity of salt hydrate PCM normally ranges between 0.2 and 0.3 W m⁻¹·K⁻¹; the low thermal conductivity of salt hydrate PCM is a drawback for high-power-density energy harvesting, and as thermal conductivity value is too low, there is a delay with the phenomenon of thermal energy charging and discharging; however, these problems do not exist with metal PCM. Based on the application to improve the thermal and chemical properties of PCM, composite additives are doped into PCM.

Overview of salt hydrate PCM

Salt hydrates are denoted as $A_xB_y \cdot nH_2O$, and n represents the number of water molecules; A_xB_y represents the type of salt (carbonate, sulphate, phosphate, nitrate and chloride). Ion-dipole bonds consisting of anion and polar molecules are the ones that are present in salt hydrate to bound salt and hydrate. Salt hydrates PCM are widely used for medium (130–250°C) and low (0–120°C) temperature heat storage applications. During the heating process, the salt hydrate desorbs the water molecule and absorbs heat, and on cooling, salt and the hydrate molecule recombine to form salt hydrate. Heating and cooling cycles are repeated continuously during heat storage application. The heat storage and heat release process of salt hydrate are shown in Figure 2(a). In addition, Figure 2(b) also represents the effect of supercooling and the nucleation phase where the solidification takes place in salt hydrate PCM during phase transition. It is evident from the graph that supercooling effect decreases the heat storage capacity and affects the heat storage performance of salt hydrate. The latent heat value of inorganic salt hydrate

PCM is 17.5% higher compared to the organic PCM, consequently, they store excess thermal energy and performs better than organic PCM [13,14].

Objective, novelty and structure of present work

The objective of this review article is to (a) provide synthesis and dispersion technique of nano additive into inorganic salt hydrate, (b) critically discuss the effect of nano additive on the thermal and chemical stability of salt hydrate PCM and (c) elaborate on the challenges and difficulties faced to commercialize salt hydrate PCM for advanced application.

A critical analysis on the performance of PCM with graphene versus the performance of PCM with boron nitrate is carried out to ensure their widespread applicability and thermophysical property enhancement [15]. A review on preparation and characterization techniques of nanoparticle dispersed phase change material (NDPCM) operating in the range of 20–37°C was also published [16]. The effect of nanoparticles in the thermal enhancement of latent heat and thermal conductivity were elaborated. Microencapsulation techniques of PCM for building cooling in terms of physical, chemical and physiochemical methods based on article published before 2019 were reviewed [17]. Another review article focused on mitigation techniques for overcoming supercooling effect in PCM and nanofluid [18]. Rathore and Shukla consolidated the benefits of macro encapsulated PCM for building cooling along with the challenges faced in macro-encapsulation [19]. Chen et al. [11] carried out a systematic review on optimization strategies of TES using PCM, mechanism of hybridizing composite PCM for energy transfer, energy conversion and advanced application using supporting materials. Aftab et al. [20] critically reviewed the current advancements and drawbacks of encapsulation techniques of PCM for TES in terms of material design. Aftab et al. [20] also summarized the application of encapsulated PCM in energy conversion and storage and in thermal rectification fields. Broad and critical roles of latent heat TES for sustainable developments in the areas of the solar thermal storage system, electro-thermal storage system and waste heat storage system were discussed [21]. Figure 3 presents you an idea about the consolidated review article available in Scopus on the term ‘salt hydrate based PCM’. Review article on salt hydrate PCM available in scopus discusses about (a) TES for greenhouse heating [22] (b) thermal property, encapsulation technique and application [23] (c) thermal stability of organic, inorganic and eutectic PCM [24] (d) encapsulation of PCM, shape stabilization and application [25] (e) clathrate hydrate PCM for cold

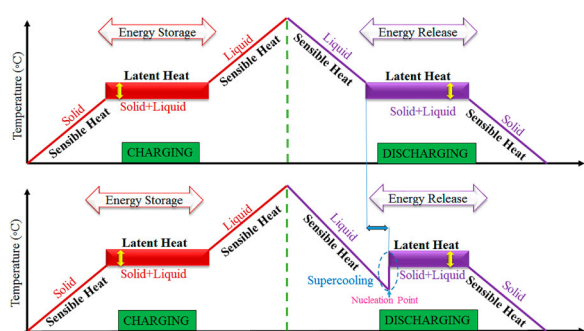


Figure 2. (a) Heat storage and release technique of salt hydrate [14] and (b) supercooling effect during charging and discharging of salt hydrate PCM.

CONTENT	YEAR	REVIEW ARTICLE ON SALT HYDRATE BASED PCM (IN SCOPUS)
Thermal Energy Storage	1998	Energy storage applications in greenhouses by means of phase change materials (PCMs): A review [22]
Thermal Property & Applications	2004	A review on phase change energy storage: Materials and applications [23]
Stability of PCM	2013	Thermal stability of phase change materials used in latent heat energy storage systems: A review [24]
PCM Encapsulation,	2014	Phase change materials for thermal energy storage [25]
Cold Energy Storage	2014	Clathrate hydrate technology for cold storage in air conditioning systems [26]
Stability of PCM	2015	Review on the methodology used in thermal stability characterization of phase change materials [27]
Thermo- Physical Property of Salt Hydrate	2016	A review of performance enhancement of PCM based latent heat storage system within the context of materials, thermal stability and compatibility [28]
PCM Melting Point 0-250 C	2016	Thermal energy storage for low and medium temperature applications using phase change materials - A review [29]
Micro Encapsulation	2017	Review of current state of research on energy storage, toxicity, health hazards and commercialization of phase changing materials [30]
Corrosion Behaviour	2017	Corrosion effect of phase change materials in solar thermal energy storage application [31]
Salt hydrate PCM for TES	2017	Inorganic salt hydrate for thermal energy storage [32]
Carbon based Organic PCM	2019	Carbon-filled organic phase-change materials for thermal energy storage: A review [33]
PCMs Melting Point 100-1680 C	2019	Medium- and high-temperature latent heat thermal energy storage: Material database, system review, and corrosivity assessment [34]
Porous Matrices PCM	2020	Clay composites for thermal energy storage: A review [35]
Enthalpy value of Salt Hydrates	2020	Critical review on measured phase transition enthalpies of salt hydrates in the context of solid-liquid phase change materials [36]
Nano Particle based Salt Hydrate	2020	Progress on use of nanoparticles in salt hydrates as phase change materials [37]
Synthesis & Hybridization Technique of Nano Additive, Enhancement & Deterioration of Thermal Property of salt hydrate, Advanced Applications of Salt Hydrate PCM		Nano Additive Enhanced Salt Hydrate Phase Change Materials for Thermal Energy Storage (PRESENT WORK)



Figure 3. Previously published a review article on salt hydrate-based PCM in Scopus.

storage application [26] (f) thermal cycling methods adopted for characterization of organic, inorganic and eutectic PCM [27] (g) thermophysical property enhancement of organic and inorganic PCM [28] (h) TES potential and application of PCM with PTT of 0–250°C [29] (i) micro encapsulation, shape stabilization and commercial application of PCM [30] (j) corrosion behaviour of PCM on integration with different application [31] (k) classification, drawback and techniques adopted to overcome the cons of salt hydrate PCM [32] (l) application of carbon-based organic PCM [33] (m) thermal property analysis of PCM with PTT 100–160°C [34] (n) porous material based encapsulation technique [35] (o) phase transition enthalpies of

chloride and nitrate salts [36] and (p) influence of nano particle in thermal properties of salt hydrate [37].

A number of review papers on salt hydrate-based PCM discuss thermal stability, nano additive dispersion for enhancement of thermal properties and various applications based on a confined range of temperature and types of salts. However, they do not focus on the synthesis and hybridization techniques of nano additive salt hydrate with enhancement and thermal stability of PCM along with the advanced application. None of these articles are comprehensively dedicated towards consolidation of research work with a focus on all major drawbacks on the application of PCM. As well as, a scientific gap exists in the

review work, where detailed information on synthesis, hybridization and preparation of composite additive-based salt hydrate PCMs are missing. Indeed a review article with a focus on additives dispersed salt hydrate PCM for overcoming the drawback of low thermal conductivity, high degree of supercooling, low corrosion resistance and lack of stability towards salt hydration for TES application would be a critical insight to scientist, researchers and engineers contributing towards clean source of energy. This review article, along with the preparation technique, provides details on enhancement and deterioration of thermal properties like melting point, latent heat and thermal conductivity of the composite additive-based salt hydrate PCM. The article also provides a significant need for commercializing nano additive PCM modules for a wide variety of TES applications.

The present review article contains five sections. General introduction on roles and dominant property of PCM, objective and novelty of the review article is presented in this introduction section. Section 'Composite additive salt hydrate PCM and hybridization techniques' reviews the hybridization technique of composite additives and the method of their doping into salt hydrate. The composite additives are classified as metal additive, nano carbon additive and polymer additive. In Section 'Advances in enhancement and deterioration of thermal properties', we critically analyse the enhancement and deterioration of thermophysical properties like melting point, latent heat and thermal conductivity of nano additive salt hydrate PCM. Numerous advanced applications of nano additive salt hydrate PCM for TES with a motto to contribute towards sustainable development goals and commercialization are consolidated in Section 'Application of composite additive salt hydrate PCM'. Conclusion and future challenges are discussed in Section 'Conclusions and future challenges'.

This review work is particularly intended for scientist, researchers and commercial manufacturers who are working in the preparation of composite additive hybridized PCM for TES. A major contribution towards SDG clean and affordable energy can be obtained by TES using PCM. TES using PCM holds application in various sectors like automotive application [38], battery thermal management system (BTMS) [39,40], building units [13,41–44], electronic gadgets [45,46], fabrics [47,48], food industry [49,50], refrigeration units, solar water heater [51–54], solar thermal system [5,55,56], space heating and cooling application [57,58] and waste heat recovery unit [59,60]. Reserves in TES along with renewable source and power efficiency, can fortify health and financial infrastructure, drive short-term improvement and align energy development with SDG.

Composite additive salt hydrate PCM and hybridization techniques

Composite additive to be dispersed in salt hydrate predominantly depends on the thermophysical property that has to be improved or modified. Dispersion of nano additive of high thermal conductivity will tend to improve the thermal conductivity of salt hydrate PCM, e.g. thermal conductivity of multi-walled carbon nano tube (MWCNT) and graphene nanoparticles is about 3000 and 5300 $\text{W m}^{-1}\cdot\text{K}^{-1}$ [61,62]. Nano additive with better corrosion inhibition properties can resolve the issue of corrosion. A better nucleating agent would tend to reduce the supercooling effect; similarly a proper binding agent will rectify the issue of phase separation. Metal additives include metals, metal oxides and metalloids like zinc, aluminium, iron, copper, titanium oxide, zinc oxide, aluminium oxide and copper foam. Carbon nano additives include carbon nano tube (CNT), nanofibres, graphene and graphite [63]. Polymer additives composites include polyaniline (PANI), polyester, polyether and polyamides [64]. Few other notable additives include ceramics and semiconductor-based composite materials. This section particularly discusses different dispersion methods, nanoparticles, proportion of nanoparticles, and the purpose of dispersing the particular nano particle with salt hydrate PCM. The mechanism of salt hydrate and nanocomposite interaction on improving the thermal conductivity of the composite mixture is shown in Figure 4 (a). Initially, on dispersion of nanocomposite over the salt hydrate PCM, the nanoparticles disperse independently over the complete surface of salt hydrate. The nanoparticles reduce the thermal interfacial resistance between the salt hydrate molecules and improve the heat flow, thereby improving the thermal conductivity of PCM, as shown in Figure 4(a). With a higher concentration of nanoparticles into PCM, they interact among themselves and form a big group of cluster, with poor distribution and form agglomeration, and hardly increases the thermal conductivity of the nanoparticles dispersed PCM. The next important problem with salt hydrate PCM is phase separation, which is solved by a form-stable structure. The intermolecular interaction between silica nanoparticles and sodium acetate trihydrate (SAT)–urea mixture is shown in Figure 4(b). Here the SiO_2 acts as a supporting material that allows salt hydrate to get dispersed over the surface and avoids phase separation during phase change. Another similar interaction between salt hydrate and EV is depicted in Figure 4(c). Figure 4(d) depicts the interaction between salt hydrate PCM and nucleating agent while suppressing the supercooling behaviour of salt hydrates. Nucleation agents increase the free energy of the nucleating surface and increase the growth of

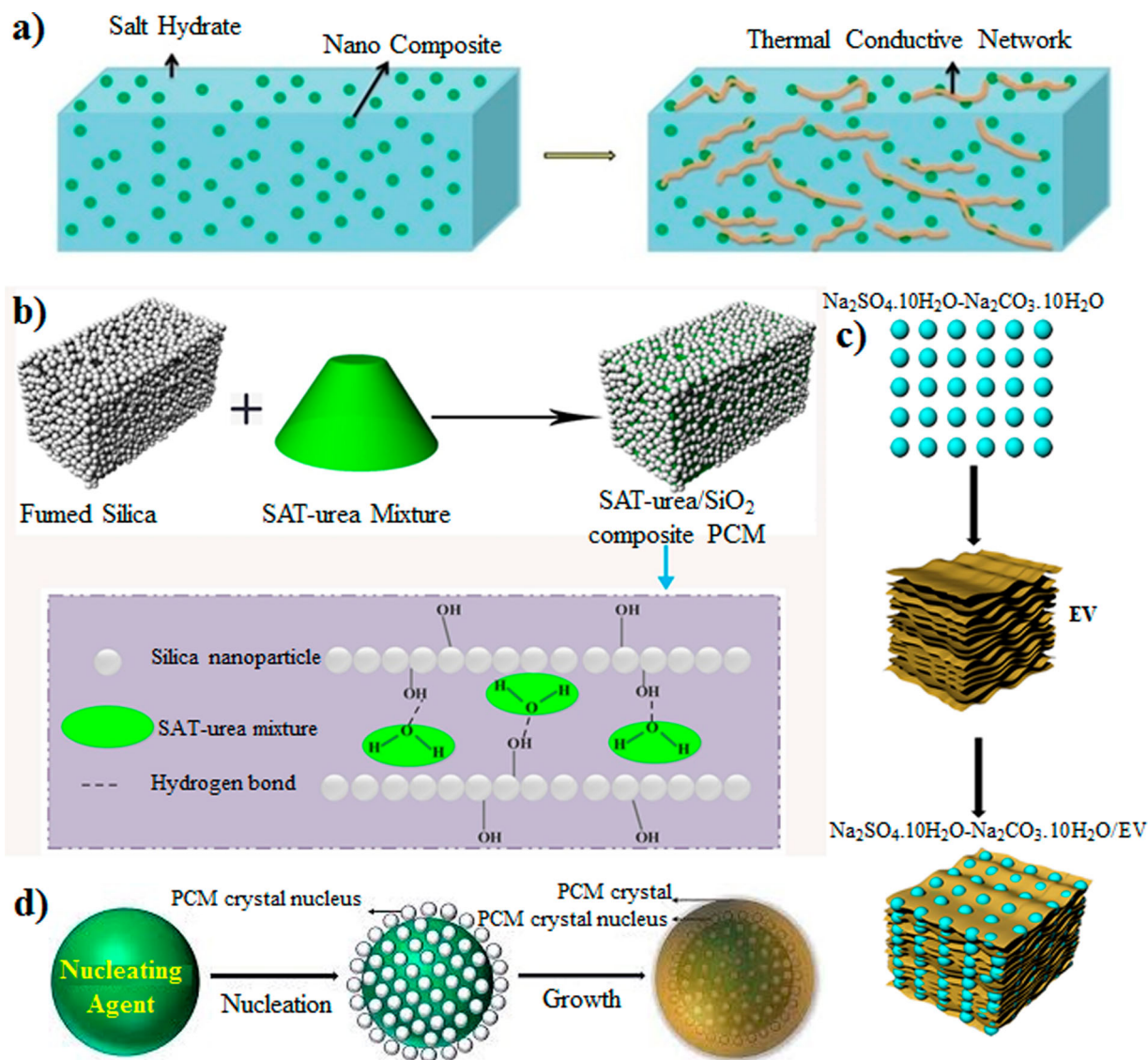


Figure 4. Interaction between salt hydrates and nanocomposite for improved (a) thermal conductivity, (b) phase stability [65] (with permission from Elsevier), (c) phase stability by foam stable [66] (with permission from Elsevier) and (d) degree of supercooling with nucleation agent [67] (with permission from Wiley).

the PCM crystal nucleus above the nucleating agent. As a result of the above interaction, the heterogeneous nucleation rate increases in salt hydrate and reduces the degree of supercooling.

This section also critically analyses the ease towards small, medium and large-scale synthesis and dispersion methods for effective commercialization. A separate section on various dispersion techniques with spectroscopic evidence to ensure hybridization of nanoparticles with salt hydrate PCM is included.

Metal additives

Metals with desirable mechanical and thermal characteristics are commonly mixed with salt hydrate PCM to enhance the thermophysical property of salt hydrate. In current trends, metal foams and oxide of metal particles are doped to improve the desirable properties. It is also noted that the inclusion of

metal additives (Ag, Cu, Pt, Al₂O₃ and SnO₂) into PCM exhibit better thermal properties and are worth using as carrier materials. Synthesis of metal additives is based on techniques like chemical co-precipitation, emulsion process or phase transition technique. Most desirable metal composite additive is metal foam as it exhibits good strength, stiffness, large surface area and high thermal conductivity. Metal foam reveals a dominant role in improving the heat conduction rate as well, and it weakened the supercooling effect by decreasing the kinetic coefficient [68].

Zhou et al. [69] used CaCl₂·6H₂O as salt hydrate PCM, and porous material copper foam as a metal additive in his experimental investigation. An attempt was carried out to address the supercooling issue of CaCl₂·6H₂O. Results show the degree of supercooling to be 5°C for salt hydrate with copper foam matrix and 8°C for salt hydrate without copper foam. Li et al. [68], in their experimental investigation, used hydrate salt

of SAT as PCM. This investigation was carried out to address the issue of low thermal conductivity, degree of supercooling and phase separation of SAT. Copper foam as supporting matrix, disodium hydrogen phosphate dodecahydrate (DSP) as nucleating agent and carboxyl methyl cellulose (CMC) as thickener were used as the additive. Initially, SAT and DSP were blended at room temperature followed by heating of salt hydrate in a water bath until 80°C. Second, CMC was added to the liquid mixture and the sample was thoroughly mixed using magnetic stirring. Modified SAT PCM was impregnated into the porous space of copper foam to enhance shape stabilization and thermal conductivity. $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ salt hydrate is another commonly used PCM for TES due to its low melting temperature of 35°C and high latent heat potential, however, the effect of supercooling has to be solved for wide applications. Kumar et al. [70] experimentally characterized the performance of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ salt hydrate by mixing nucleation agent. Zinc oxide was included as a nucleator by ultrasonication process to resolve the supercooling issue. The effect of zinc oxide in addressing the degree of supercooling was studied for 1000 thermal cycles, and it was noted that the modified salt-hydrated effectively stores and releases heat energy up to 750 cycles. Raj et al. [71] performed a characterization study using manganese organo-metallic salt hydrate (MSS-PCM). In his research work, he studied the chemical stability effect of metal gallium on MSS-PCM after different thermal cycles. Initially, metal gallium was encapsulated into pill in weight fraction of 5% and 1%. The synthesized nano metal gallium pills were dispersed into MSS-PCM to analyse their characteristics. Figure 5(a) depicts the encapsulation technique adopted for preparing nano metal gallium pills. The synthesis technique of MSS-PCM using magnetic stirring is shown in Figure 5(b). Figures 4(d)–5(c) show the photograph of hybrid PCM in wt-% of 95–05% before use and after 1000 number of thermal cycles. Scanning electron microscope (SEM) image in Figures 4(g)–5(e) shows the morphological structure and the presence of gallium pills in MSS-PCM of weight fraction 95–05%.

All dispersion and hybridization of metal additive with salt hydrate PCM are carried out by a combination of techniques like (a) blending and melting, (b) melting and impregnation and (c) melting and magnetic stirring followed by ultrasonication. The author would like to comment that, in all hybridization techniques, uniform dispersion of composite additive occurs when they are added to the liquid state PCM. And there is always a need for a dual technique for hybridization. It has been observed from the above-discussed research works that metal additives are actively used with salt hydrate PCM for enhancing thermal property and thermal

stabilization. Nevertheless, none of the above research works discusses the technique to hybridize metal additives for large-scale application or for industry. All synthesis, hybridization technique, testing and characterization are designed only for a lab scale. Another major point that is not discussed by the authors is the corrosion effect of salt hydrate on impregnating with metal foams. Though metal foam offers better shape stabilization and enhanced thermal properties, the corrosive nature of salt hydrate with metal need to be discussed. For the above issue in regard to corrosion, plenty of research works are carried out by blending metal additive with organic PCM rather than inorganic salt hydrate PCM. Table 1 distinctively comprises all the research works carried out in the last decade on doping of metal additive into salt hydrate. Details on the concentration of metal additives doped, the purpose of the additive and the performance enhancement of salt hydrate due to doping of metal additives are discussed.

Carbon additives

Variety of carbon nano additives is widely used with salt hydrates to overcome the problem of low thermal conductivity. The conductive nature of carbon nanoparticles is highly responsible for the thermal conductivity of the composite on hybridization. In addition, the thermal conductivity of salt hydrate is greatly influenced by the type of carbon material, dimension of carbon material, surface area of the nano additive and concentration of doping. CNT, CNF (carbon nano fibre), graphene and graphite are the commonly doped carbon additive with salt hydrate PCM. Thermal property enhancement depends on the surface area of additive; however, graphene and single-walled CNT exhibit high thermal properties due to higher surface energy and surface area. As well, the inclusion of carbon additives make a negative impact on the energy storage density of the composite, as the inclusion of nano additives decreases the amount of active salt hydrate [75]. 2D carbon nanoparticles also act as nucleation agents and contribute towards suppressing of supercooling effect in the salt hydrate. Figure 6 consolidates the classification of carbon additives based on the dimension and also provides insight into their shape, surface area and thermal conductivity. Depending on the thermal property and structure, the author has rated each type of additives for doping with salt hydrates.

Zero dimensional (fullerenes and carbon dots)

Closed convexity molecules formed by carbon atoms are fullerenes. Fullerene groups are made of pentagonal and hexagonal faces using carbon atoms based on C_n cluster, where n value is greater than 20 [76].

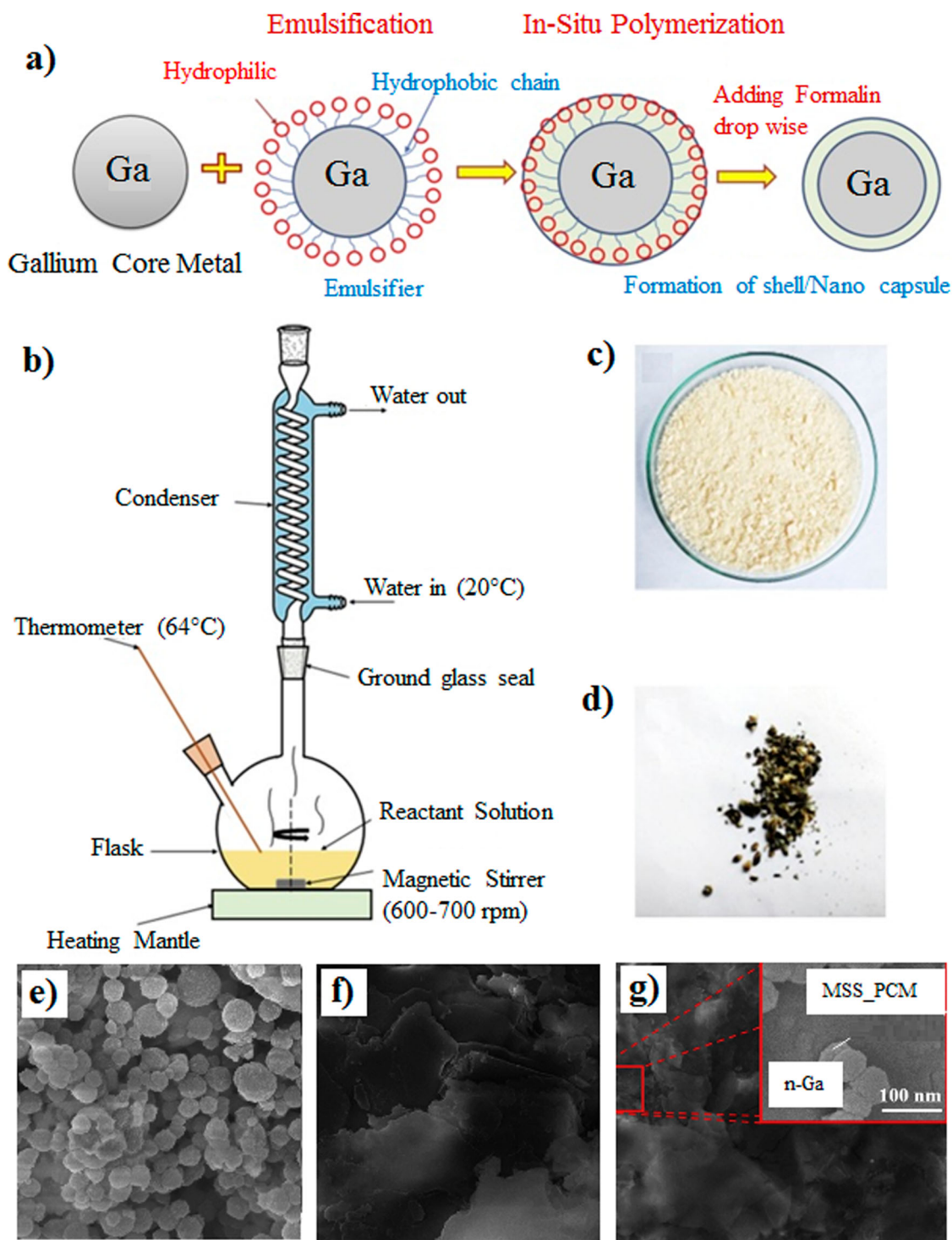


Figure 5. (a) Nano encapsulation technique of metal gallium, (b) hybridization method of doping encapsulated gallium into MSS-PCM at the composition of MSS-PCM/n-gallium in wt-% 95:05 and 99:01, (c) MSS-PCM before use at 40°C, (d) MSS-PCM after use at 130°C, (e) field emission scanning electron microscope image of nano encapsulated gallium, (f) SEM image of MSS-PCM and (g) SEM image of MSS-PCM/n-gallium in wt-% of 95:05 [71] (with permission from Elsevier).

Owing to the unique spherical structure and due to the ease of modification of structure in the molecular-level fullerene are extremely used in energy-related applications like energy transformation, energy storage and photo electronics [77]. Fullerene is electron affluent carbon material, thus it exhibits good electronic conductivity, strong structure and good chemical stability. Fullerenes are actively used in super capacitors [78] for storing electrical energy as well

fullerenes are used as electro catalysis in fuel cells [79] for satisfying the energy demand. However, due to the low thermal conductivity of fullerene, they still lag their footprint in TES applications and are yet to be explored with salt hydrate PCMs.

One dimensional (CNTs/CNFs)

CNT are well arranged hollow cylindrical nanostructures of carbon atom. CNT exhibit outstanding

Table 1. Summary on metal additives based salt hydrate.

Ref.	Salt hydrate	Metal additive	Conc. (%)	Purpose	Performance
[68]	CH ₃ COONa·3H ₂ O	Copper foam	0.5 1.5 05	To improve thermal conductivity	Thermal conductivity of copper foam/SAT increase by 11 times
[69]	CaCl ₂ ·6H ₂ O	Copper metal foam	–	To enhance heat transfer rate	Heat transfer rate of salt hydrate with copper metal foam increases by three times
[70]	Zn(NO ₃) ₂ ·6H ₂ O	(a)ZnO (b)Zn ₃ (OH) ₄ (NO ₃) ₂	05	To suppress degree of supercooling	Zinc hydroxyl nitrate suppresses degree of supercooling of salt hydrate reduced from 12°C to 3°C for 750 thermal cycles
[71]	Manganese organo-metallic salt hydrate	Gallium	01 05	To improve thermal conductivity and enthalpy	Thermal conductivity increased by 26.8% and enthalpy increased by 13.5%
[72]	9 parts of Na ₂ HPO ₄ ·12H ₂ O + 1 part of Na ₂ SO ₄ ·10H ₂ O	2-(2-Pentadecyl-4, 5-dihydro-1H-imidazol-1-yl) ethanol (PDIE)	–	To reduce corrosion of Cu/salt hydrate interface	PDIE acts as a corrosion inhibitor at copper metal/PCM surface. The corrosion inhibition mechanism is also discussed
[73]	Na ₂ SO ₄ ·10H ₂ O (43.2% PCM) Na ₂ HPO ₄ ·12H ₂ O (12.3% PCM) NH ₄ Cl (6.2% depressant) KCl (3.1% depressant) NH ₄ H ₂ PO ₄ (3.1% buffer) H ₂ O (27.8% Solvent)	Borax (1.3%) CMC (2.5%) Methionine and valine	–	To study the corrosion reaction of metal material with salt hydrate in the presence of methionine and valine	Methionine and valine acts as an effective corrosion inhibitor for carbon steel
[74]	Na ₂ S ₂ O ₃ · 5H ₂ O	Aluminium foam	–	To enhance thermal conductivity and shape stability	Thermal conductivity increases from 0.45 to 10.8 W m ⁻¹ ·K ⁻¹ with enhance foam stability

mechanical, electrical and thermal properties due to their surface area, thermal conductivity and morphology [76]. They are basically categorized based on structure as (a) single-walled carbon nano tube (SWCNT) (b) double-walled carbon nano tube (DWCNT) and (c) MWCNT. The property of CNT varies with respect to the degree of chirality and diameter of the nanotubes. CNF are formed by graphene nano sheets. It has been observed using transmission electron microscope (TEM) that the graphene layers are not arranged along the axis of fibre, and depending on their angle of arrangement, CNF are classified as graphene layers (a) perpendicular to fibre axis (b) piled at angle to the fibre axis. CNF are inorganic materials and about 90% of their composition is carbon element. Inorganic material with carbon elements above 99% is graphite fibre. The length and diameter of CNF are about 100 µm and 2 nm. Generally, the microstructure of nanomaterials influences the thermochemical properties of PCM. Doping nanoparticles, nanotubes as well as nanofibres improves the thermal characteristics of colloidal suspension PCM [80]. KNO₃ molten salts are doped with nanoparticle oxide like Al₂O₃ and SiO₂ of size 2–100 nm; however, the PTT of the colloidal suspension PCM remains unaltered. Rabady and Malkawi [81] experimentally evaluated the performance of sodium thiosulphate pentahydrate (Na₂S₂O₃·5H₂O) with CNT nano additive. Salt hydrate PCM was melted till it attains liquid phase using a water beaker on a hot plate. A measured proportion of CNT was added into salt hydrate and

was thoroughly mixed using a magnetic stirrer followed by sonication for 50 min, in addition to the nano additives; nucleation agent CMC and sodium dodecyl sulphate as a surfactant was mixed with the composite. The molten mixture of composite PCM was made into a rectangular plate using a hydraulic plate for analysing these thermal properties. Based on the characterization results it is observed that the composite salt hydrate has higher thermal conductivity compared to the base salt. CNT effectively decreases the charging and discharging time for heat storage.

Two dimensional (graphene)

Graphene are made of a hexagonal chain of carbon layer with adjacent carbon atoms connected strongly using a covalent bond. Thickness of graphene atom is equal to one carbon atom as it is a two-dimensional material with high stability. Owing to their inimitable molecular-level structure, graphene possesses thermal conductivity about 3000–5000 W m⁻¹·K⁻¹ and this value is quite higher than many metallic elements. Thus, doping of graphene material into the networks of salt hydrate PCM would effectively improve the thermal conductivity for TES. Nano sheets of graphene molecule improve the stability of salt hydrate and contribute to overcoming the issue of phase separation; however, the thermal conductivity of salt hydrate on doping nano sheets of graphene once again depends on molecular structure.

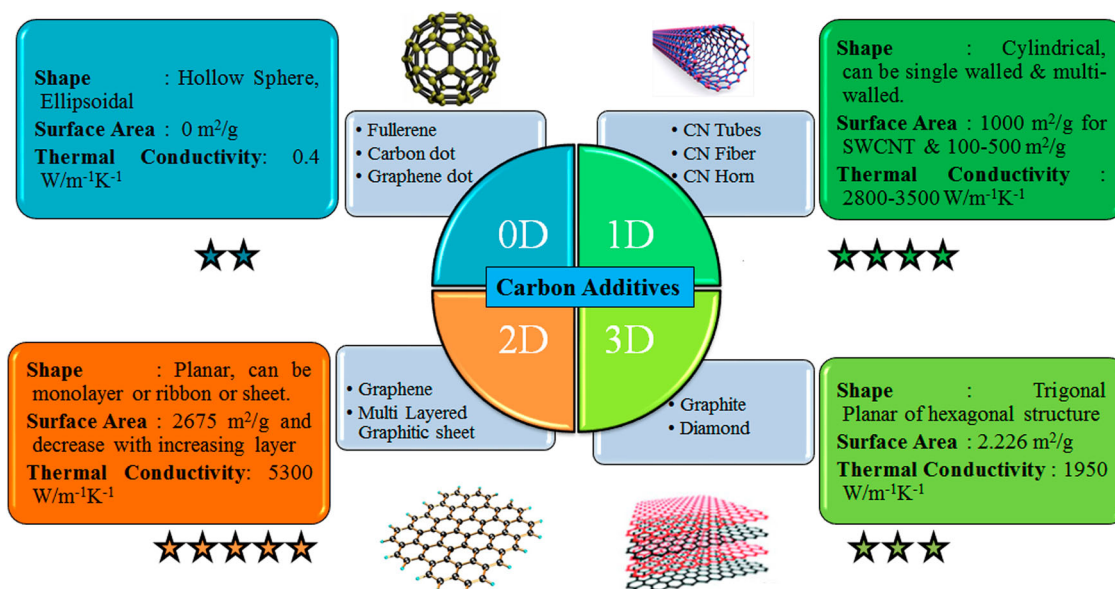


Figure 6. Classifications of carbon nano additives based on dimension.

Wang et al. [82] prepared a foam stable PCM using Fe_3O_4 and graphene nano sheets. The composite preparation mechanism is well illustrated in Figure 7(a). Results show the prepared composite to exhibit outstanding thermal stability with high PTT and effectively absorbs sunlight for better heat storage. Liu et al. [83] investigated on thermal management of salt-hydrated hydrogels using 2D carbon nanoparticle. Modified hydrate salt (EHS) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ and $\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$ were used as PCM. GO and polyacrylamide-co-acrylic-acid (PAAAM) were used as carbon additive and copolymer hydrogel. The phase transition behaviour of $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ and $\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$ was studied by the interaction between GO and PAAAM. The foam stable co-polymer hydrogel salt was prepared to address the phase separation issue. The unique preparation technique and the mechanism of PAAAM and GO binding with salt hydrate are explained in Figure 7(b). Results show 54% improvement in thermal conductivity of GO-EHS/PAAAM compared to EHS/PAAAM composite. The copolymer hydrogels contribute towards adjustable phase change temperature of the inorganic salt hydrate. By product of many chemical processes is $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$, and it possesses a latent heat capacity of 209 J g^{-1} with a melting point of 29°C ; however, they need to overcome the supercooling effect to be efficiently used for TES. Xu et al. [84] developed a novel composite PCM for building heating applications. Calcium chloride hexahydrate $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ was used as salt hydrate PCM, nano sheets of GO as nano additive at 0.2, 0.5 and 0.8 wt-% along with $\text{SrCl}_2 \cdot 6\text{H}_2\text{O}$ as nucleation agent for suppressing degree of supercooling. 50 g of CaCl_2 salt was initially mixed with deionized water in a water bath at heated till 40°C to ensure liquid phase, now graphene nano additives

are dispersed. In order to avoid agglomeration of nano additives, mixing was carried out using ultrasonicator. Zhang et al. [85] carried out his research investigation using $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ as salt hydrate PCM. A new nano composite salt hydrate r-GO/ $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ was prepared by doping reduced GO. The investigation was focused on phase-change enthalpy, thermal reliability and thermal conductivity of the composite prepared. Hydrate salts were doped into the aqueous graphene followed by stirring and mixing using ultrasonicator for 30 min. An elaborated schematic view of the nano composite salt hydrate is shown in Figure 7 (d), and it also depicts the phase change enthalpy of $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ with nano additives during crystallization and melting. Results depict improvement in thermal conductivity of the composite by 80% with 2.7% drop in latent heat. The composite was tested for 100 thermal cycles and noticed to have drop in latent heat of composite with a significant limit. However, in real-time application, the PCM is considered to be economically beneficial only when it offers reliable property for a higher number of thermal cycles.

Thermochemical materials (TCM) are one that has greater potential to store energy by reversible chemical reaction without any thermal losses. For medium-temperature TES applications, new TCM were synthesized by mixing MgSO_4 and MgCl_2 with graphene sheets [87]. Three different TCM composite (a) graphene- $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, (b) graphene- $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ and (c) graphene- $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ / $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ were synthesized by impregnation of hydrated salt within graphene sheets. Owing to latent heat value of $130\text{--}150 \text{ kJ} \cdot \text{kg}^{-1}$ and low-cost magnesium nitrate hexahydrate $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ make themselves highly potential to be used in buildings. Wang et al. [88] experimentally studied the

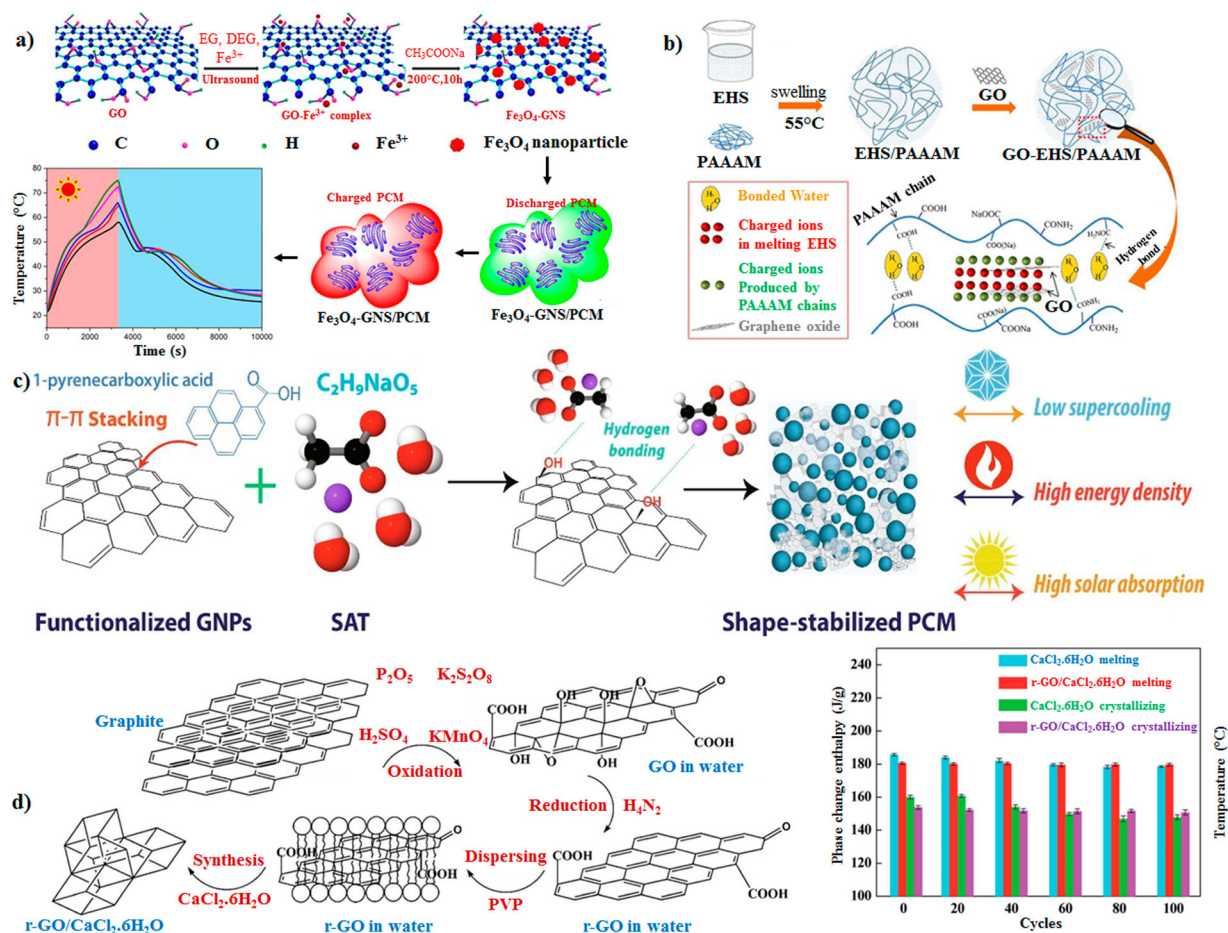


Figure 7. Synthesis technique of graphene-based salt hydrate. (a) Formation mechanism and thermal energy storage in Fe_3O_4 – GNS composite [82] (with permission from RSC), (b) synthesis and bonding mechanism of GO-EHS/PAAM [83] (with permission from RSC), (c) SAT hybridized with graphene nanoplatelets to suppress supercooling effect [86] (with permission from Elsevier) and (d) hybridization technique of doping r-GO with $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ and the composite G/ $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ and r-GO/ $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ composite PCM [85] (with permission from RSC).

thermal performance of $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ by incorporating graphene at 1, 3 and 5 wt-%. With graphene as a nano additive, CMC was doped as a surfactant. The mixing was carried out by sonication after dispersing carbon nano additive with salt hydrate in the liquid phase. The problem of poor hydrophilia of carbon materials especially multi-dimensional carbon nano particles with salt hydrate PCM are a notable issue to rectify. Mehrali et al. [86] carried out an experimental characterization study on SAT PCM with graphene nanoplatelets (GNP) as nano additive. This particular work was carried out to suppress the degree of supercooling and to improve the thermal conductivity of the salt hydrate PCM. It's been noted that doping of functionalized GNP contributes towards uniform dispersion of nano additives on the salt hydrate by enhanced hydrophilic nature. Figure 7(c) shows the mechanism between the hybridization of SAT and GNP. As a benefit in addition to improvement of thermal conductivity of SAT by 114%, GNP also prevents leakage and enhances the stable stabilization of carbon additive PCM. Shen et al. [89] experimentally investigated

to enhance the TES performance of SAT PCM by mixing cellulose nanofibril and GNP as additives. The major issue with SAT is phase separation and low thermal conductivity. In this investigation, along with nano additives, $\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$ was doped as a nucleation agent. Dispersion of graphene composite, DSP and SAT in deionized water was performed using magnetic stirring for 1.5 h. Initially, cellulose nanofibril was mixed with salt hydrate in wt-% of 0.0%, 0.2%, 0.4%, 0.6%, 0.8% and 1.0% to determine the best suitable wt-% opted to reduce phase separation issue of SAT. Figure 8 illustrates the hydrophobic interaction with SAT and the additive; it also shows the SEM image with the dispersion of CNF and GNP over SAT. Results confirmed 0.8% cellulose nanofibril increases the elastic modulus of the SAT PCM and provides a stronger network to eradicate phase separation. Followed by CNF graphene nanoplatelets were added in 0.5%, 1.5% and 2.5%. GNP disperse CNF uniformly in the base salt hydrate and improves the thermal conductivity from 0.442 to $0.686 \text{ W m}^{-1} \cdot \text{K}^{-1}$ for the GNP doped SAT. The thermal cycling test shows a drop

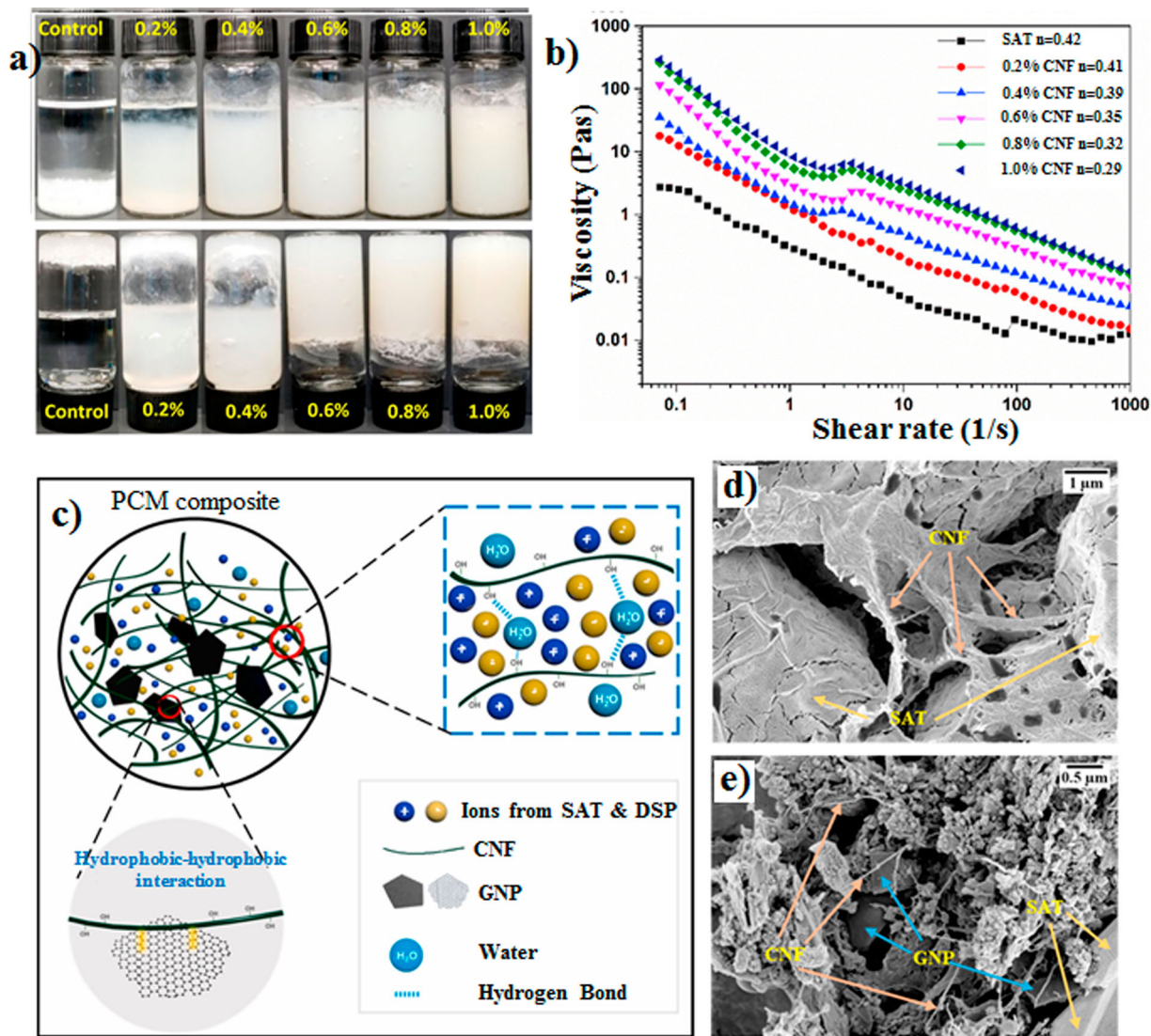


Figure 8. (a) Phase stability and fluidity of SAT with different proportion of CNF, (b) viscosity of SAT/CNF composite, (c) hydrophobic interaction between ions of SAT, DSP, CNF and GNP, (d) SEM image of CNF dispersed SAT and (e) SEM Image of CNF/GNP dispersed SAT PCM [89] (with permission from Elsevier).

in enthalpy and melting point of the mixture after 100 cycles; on the contrary, the nano additives facilitate better resistance to phase separation.

Different techniques of hybridization and dispersion of graphene over the base salt hydrate PCM are preferred by various researchers. However, none of them discuss the technique for doping carbon nano additive with salt hydrate PCM for the industrial level. And there is still a need for economic analysis of the composite prepared for their commercialization. Thermal reliability and thermal stability of nano additive PCM need to be evaluated for a higher number of thermal cycles. A numerical modelling in regard to the nanostructure doped PCM and their thermal property is missing.

Three dimensional (graphite)

Graphite is made of carbon atoms arranged in the hexagonal crystal structure, and they appear dark grey in colour. Nano graphite has unique attributes like (a)

easy synthesis compared to graphene materials; (b) better electron mobility that improves the electrical and thermal characteristics; (c) superior mechanical strength [90] to be used as nano filler with salt hydrate PCMs. Using sonication technique, graphite nanoplatelets are easily synthesized using expanded graphite by modifying its structure. Galazutdinova et al. [91] prepared and characterized the performance of 40% $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ and 60% $\text{MgNO}_3 \cdot 6\text{H}_2\text{O}$ eutectic inorganic salt hydrate with different preparation techniques. Expanded graphite (EG) flakes and EG matrix were used as carbon nano additives to address the low heat transfer rate and to ensure shape stabilization. Three different techniques were adopted for hybridization of each nano additive, like EG flakes and EG matrix with salt hydrate. Thermal property like PTT, specific heat, latent heat of fusion, latent heat of solidification and degree of supercooling of the composite PCM was characterized. (a) 5%, 10% and 15% EG flakes were added to the eutectic salt

hydrate by physical mixing, (b) 5% EG flakes were added to the eutectic salt hydrate by ultrasonication and (c) 10% and 15% EG flakes were added to the eutectic salt hydrate by impregnation. In case of EG matrix rather than the percentage of nano additive, the time of preparation was varied. Salt hydrate and EG matrix composite were prepared using the technique (a) vacuum impregnation, (b) immersion and (c) ultrasonication. In regard to EG matrix, the optimum value of ΔH ratio is found to be with sample preparation time of 1 h for vacuum impregnation, 3 h for ultrasound and 4 h for immersion method. EG matrix also supports in suppressing the degree of supercooling of eutectic salt hydrate without the presence of a nucleating agent. With respect to EG flakes with salt hydrate, there were no satisfying data obtained with respect to differential scanning calorimetry (DSC) and thermal gravimetric analyser (TGA). However, the performance of EG flakes with 40% $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ and 60% $\text{MgNO}_3 \cdot 6\text{H}_2\text{O}$ salt hydrate needs further experimental study. $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ is an assuring salt hydrate PCM with plenty of availability from the lake. In another research work, Zhou et al. [92] prepared and investigated the thermal performance of composite PCM using $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ as the base inorganic salt hydrate. EG was modified initially to improve its hydrophilicity using triton X as a surfactant. The modified EG was used as a nano additive. This particular investigation was carried out to address the leakage issue of salt hydrate PCM during phase transition and to suppress the degree of supercooling. The composite samples were prepared by drying, ultrasonication and impregnation techniques. Initially, EG powder was dried in an oven for 12 h followed by its dispersion in triton solution through the sonication process. The sample is dried at 80°C to obtain modified EG powder, which is later compressed into round block. Modified EG round blocks were immersed in a solution of $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ to obtain the composite PCM. Results showed 2.7% of drop in latent heat of composite material developed after 500 thermal cycles, whereas the drop in latent heat was about 41.7% with pure $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ salt hydrate. As well, the degree of supercooling of the composite PCM dropped to 1.3°C from 11.2°C for pure salt hydrate.

Zou et al. [93] conducted a study to understand the cold energy storage potential of $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ salt hydrate. Graphite of three different sizes 50, 80 and 100 mesh were mixed as nano additive. This meticulous study was carried out to address the issue of low thermal conductivity and supercooling effect. The impregnation technique was adopted to prepare the $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ /EG composite. The shape stabilization of modified salt hydrate with graphite of 50 mesh sizes was superior due to the presence of maximum micro pores and pore volume, compared to the

graphite with 80 and 100 mesh sizes. Similarly, the thermal conductivity and latent heat of composite PCM with graphite of 50 mesh sizes were excellent due to low thermal resistance and minute captivity effects. The effect of supercooling was also reduced with graphite as a nano additive. Bao et al. [94] developed a stable inorganic PCM for reducing the heat fluctuation caused by solar radiation. Flake graphite is used as a nucleating agent, and absorbent polymer is used as a thickening agent in this investigation. The issue caused due to the degree of supercooling and phase segregation was expected to reduce in developing this inorganic composite PCM. Different proportions (0.2, 0.5 and 0.8 wt-%) of flake graphite were mixed to 50 g of $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ in a beaker through a magnetic stirring process followed by ultrasonic dispersion for 30 min. In order to mitigate the phase separation effect, a thickening agent was included by stirring at $1500 \text{ rev min}^{-1}$. The degree of supercooling for salt hydrate with 0.2, 0.5 and 0.8 wt-% of flake graphite were 1.7°C , 0.9°C and 2.6°C correspondingly. Effect of flake graphite as a nucleating agent, not only reduces the degree of supercooling but also improves the thermal conductivity of the composite PCM due to the conductivity nature of graphite.

Wu et al. [95] study the performance of salt hydrate PCM for solar energy storage numerically and experimentally. In his research work, he opted aluminium potassium sulphate dodecahydrate $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ as inorganic salt hydrate PCM. EG was added as an additive in five different proportions of 10%, 15%, 20%, 25% and 30% to prepare a form-stable PCM. The major issues like degree of supercooling, low thermal reliability, low thermal conductivity and phase separation are addressed. To resolve the leakage of water molecules from the pores of EG during charging and discharging of thermal energy, the technique of hydrophobic and hydrophilic is proposed through the surface treatment of EG lamellar surface using a surfactant. Modified aluminium sulphate dodecahydrate was synthesized by adding CMC and $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$. The solution of modified PCM salt hydrate is poured into the beaker of melted EG solution and stirred thoroughly using a glass later. Later the samples are heated in a vacuum drying box for one day and are later made into form-stable pressed tablets. The PCM preparation process involved melting, blending and solidification processes. The preparation process of composite PCM involves many steps and no specific reasons are provided for the hybridization technique, which leaves no information for the better understanding of the readers.

Li et al. [96] work was focused on enriching the application of inorganic salt hydrate for building. $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ was used as PCM, EG was opted as carbon additive, CMC and octyl phenol polyoxyethylene ether was selected as a binding agent. The major concern

was to address phase separation and supercooling. $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ and CMC mixture were vigorously stirred in a water bath for 30 min to ensure complete melting, later via vacuum impregnation technique EG were included to stability of salt hydrate PCM. EG was added at a proportion of 5, 6 and 7 wt-% to study the supercooling and phase separation effect. Results show 7% EG to reduce degree of supercooling by more than 19°C , with 3.6 times higher thermal conductivity. Rabady and Malkawi [81] also evaluated the thermal performance of sodium thiosulphate pentahydrate $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ with graphite nano additive at 1%, 3%, 5% and 7% of the mass fraction. CMC and sodium dodecyl sulphate were used as a nucleating agent and surfactant. This investigation was carried out to address the lower thermal conductivity of $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$. Salt hydrate PCM was melted till it attains liquid phase using a water beaker on a hot plate. A measured proportion of graphite nano powder was added into salt hydrate and was thoroughly mixed using a magnetic stirrer followed by sonication for 50 min, in addition to the nano additives; nucleation agent CMC and sodium dodecyl sulphate as a surfactant was mixed with the composite. Thermal conductivity of composite salt hydrate was enhanced by 155% using graphite nano particle of 7% mass fraction.

Salviati et al. [97] performed an experimental analysis with a focus on improving the thermochemical energy storage potential in hydrate salt. $\text{SrBr}_2 \cdot 6\text{H}_2\text{O}$ was used as inorganic salt hydrate PCM. Expanded natural graphite nano composite was used as an additive. The aim of this work was to ensure better stability of hydrate salt with composite PCM by enhancing the graphite interfacial surface and heat transfer rate during phase transition. The wet impregnation method was followed to prepare the powder composite and freeze-drying was adopted to prepare the porous structured composite as shown in Figure 9(a). Samples were chosen in the fraction of 5:1:1 (salt hydrate/graphite/cellulose nano fibrils). The synthesis technique involved the suspension of cellulose nano fibril in water to form binder solution, followed by mixing of graphite. The solution was magnetically stirred to ensure proper dispersion of graphite without agglomeration. Later SrBr_2 salts were added and impregnation technique was carried out to remove water. The presence of binder improved the hybridization of graphite with salt and provided a mechanical stability and graphite improved the thermochemical properties for energy storage application. A modern salt hydrate PCM was prepared by Jin et al. [98] by using SAT, potassium chloride KCl, urea and expanded graphite material. First, the 8 wt-% KCl and 3 wt-% urea were added to SAT salt hydrate by melting and blending technique. The composites were magnetically stirred for 10 min to ensure uniform distribution of composite, later by vigorous

stirring graphite's were mixture at different weight ratio of 3%, 06%, 09% and 12%. Figure 9(b) depicts the preparation technique of composite PCM SAT-KCl-urea/EG. The prepared composite PCM exhibits 1.56°C supercooling after 100 thermal cycles.

Many research works carried out discusses enhancing the thermal performance of salt hydrate PCM; however, there is a lack of experimental investigation and characterization on the cold thermal energy storage performance of inorganic salt hydrate with nano additives. Cold thermal energy storage is a process that involves adding cold thermal energy to a medium and extracting it whenever it is needed. During the charging process, the available cold thermal energy can be accumulated into the storage medium. In addition, there is a lack of studies about low melting point PCM. And in many works, there are different hybridization techniques which follow both aqueous and dry mixing methods, but no proper discussion and reasons are provided for the methods opted. For using composite PCM in industries, it is required to have a common hybridization technique, which can be opted universally. Strong affinity towards water molecule is a much-required phenomenon while doping carbon additives with salt hydrate PCM. On the contrary, they exhibit poor hydrophilia with salt hydrate PCM, especially multi-dimension carbon materials. Recent research works involve the addition of hydrophilic macromolecules onto the CNT additives; this modifies the surface of carbon material and develops polar characteristic and makes them easy to disperse with PCM [100]. This modification develops an affinity to bind between the polar CNT and organic/inorganic PCM. The surface of CNT fillers can also be modified by heat treatment. During heat treatment, there is a formation of defects and functional groups over the surface of CNT. Heat treatment transfers CNT from hydrophobic to hydrophilic and enhances the interface between CNT and PCM [101]. Numerous research works are carried out to study the dispersion of CNT, SWCNT and MWCNT with aqueous organic and polymer solution. There is a lack of information on the poor hydrophilia of carbon material with inorganic salt hydrate PCM.

Polymer additives

Polymer nano composite (PNC) is a matrix, reinforced by various nanofiller having long-chain molecules formed by repeated identical structures [102]. PNC has exclusive physicochemical properties like (a) light in weight, (b) superior thermal property, (c) flammability property, (d) anticorrosive nature and (e) shape stabilization potential. They enhance the property of the base material when loaded in a small quantity. Due to the anticorrosive properties of polymer, they are predominantly used as an anticorrosive

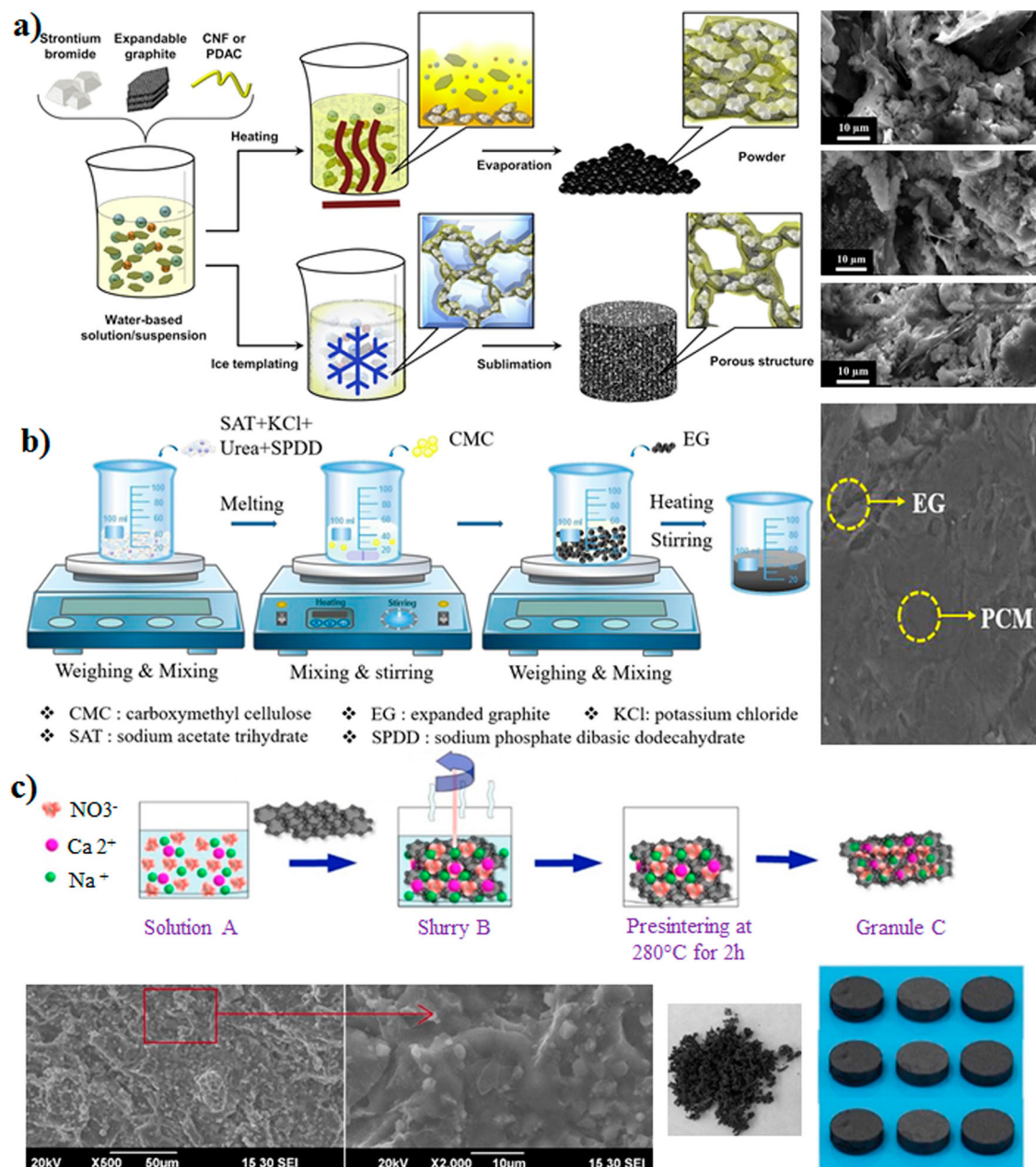


Figure 9. Synthesis and hybridization technique of doping graphite additive into salt hydrate. (a) Preparation technique of thermochemical material composite with SEM image [97] (with permission from Elsevier), (b) hybridization technique of SAT/EG/KCl composite PCM along with its SEM image [98] (with permission from Elsevier) and (c) synthesis technique of $\text{Ca}(\text{NO}_3)_2 \cdot \text{NaNO}_3/\text{EGSS-PCM}$ and its SEM image and real image of EG [99] (with permission from Elsevier).

coating layer. Typically polymers are grouped based on their structure as homochain and heterochain, based on processing characteristics as thermosets or thermoplastic and on their polymerization mechanism as addition polymers or condensation polymers. PNCs are widely adopted for environmental remediation, energy storage applications, defense system, information technology sectors and as novel catalyst. A new variant of polymer are conducting polymer, which display wide series of electrical conductivity, mechanical flexibility and thermal stability. Few conducting polymer and their chemical structures are shown in Figure 10, these conducting polymers are doped with organic and inorganic PCMs to improve

thermal conductivity and to enhance shape stabilization. Depending on the level of doping and the nature of the dopant, the polymers can be conductors, semi-conductors or insulators. Similarly, based on the type of doping and chemical reactions, the electrical property of conducting polymers can be improved drastically. Nano fillers are incorporated with polymers to enhance various distinct mechanical properties. Uniform distribution of nanofiller over polymer develops strong interfacial interaction between the matrix and improves the mechanical properties. PANI is a conducting polymer and a metallic compound, hence it was doped as a polymer additive with PCM to improve the thermal conductivity of PCM [103]. Effect on

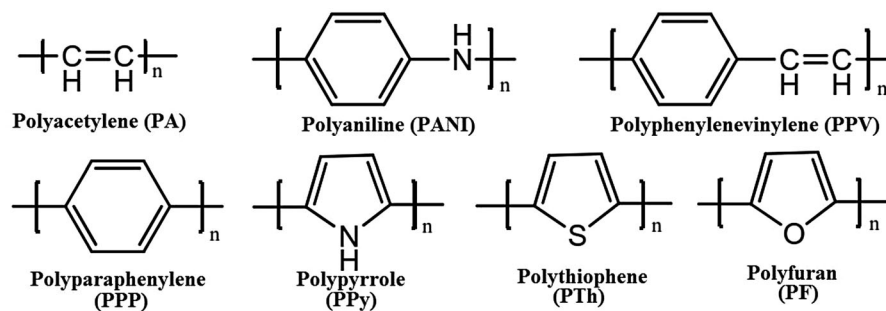


Figure 10. Chemical structure of conducting polymers [102] (with permission from RSC).

doping PANI with organic PCM was investigated to analyse the infrared emissivity [104], foam stability [105] and thermal conductivity [106]. Results indicate that PANI has effective restraining property and reduces the emissivity of infrared rays, they also show good shape stability and do not allow the PCM to liquefy. Due to the conductive nature of PANI, they form thermal networks on dispersion with PCM and supports electron thermal transport and increase the thermal conductive nature of the composite. However, on excess concentration PANI nano additive gets agglomerated and disturbs the thermal networks. There are only a few remarkable work on doping of polymer additive with salt hydrate PCM to improve the thermal property, however, polymer are effective used for shape stabilization and are critically reviewed.

Continuous use of inorganic salt hydrate PCM causes a problem of phase separation, which can be addressed by encapsulation and grafting copolymerization technique. Grafting PCM in the polymer structures by the support of a covalent bond is the main method to improve shape stabilization [107]. Organic polymers are used as a shell supporting material for encapsulation of salt hydrate, which supports to overcome the leakage issue and enhances the thermo-mechanical property of salt hydrate PCM. Encapsulation technique extends the application of salt hydrate PCM in textile and for medical treatments. In addition, micro porous polymers are thermally conductive in nature and are used as an additive to improve the thermal property of PCM. $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ salt hydrate has a latent heat capacity of 250 J g^{-1} and are highly capable of being used for storing thermal energy [108]. However, major issues like super-cooling and phase separation have to be resolved to improve their practical application. Yamashita et al. [108] carried out a research work by mixing $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ with different polymers with the potential to dissolve in water, the prepared samples were encapsulated using SiO_2 . To further improve the stability of the mixture, surfactant and thickener were added to the mixture. The result ensured polyethylene glycol to enhance the heat storage potential. With an aim to overcome the leakage problem and dehydration

issues in $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ salt hydrate during phase transformation, Xie et al. [109] proposed a two-step synthesise method of preparation. The salt hydrate was coated with polyurethane acrylate polymer additive to strengthen the stability. Thermal characteristic results ensured improved thermal property with high thermal constancy. The incongruent melting of SAT was prevented by doping polymer additive (a) one part of methacrylic acid (b) two parts of methyl methacrylate at lower concentration [110]. The polymer doped with SAT modifies the morphology of SAT and suppresses the incongruent melting.

Hybridization techniques and spectroscopic evidence

Hybridization is the inclusion of two nanoparticles with base PCM. Two common categories under which the hybridization of nanoparticles and salt hydrate settle are aqueous stirring and dry blending. In aqueous stirring method, the salt hydrate PCM are heated up to their melting point to transform the PCM to their liquid state followed by dispersion of nanoparticles, whereas in dry blending, the salt hydrates and nanoparticles are mixed in their solid-state. Under aqueous stirring different methods like (a) classical sonication, (b) sonication microprobe, (c) magnetic stirring, (d) impregnation and (e) mechanical agitation are included. Similarly, under dry blending, different techniques like (a) hand-mix, (b) planetary ball mill and (c) freezer/mill are categorized. Few recent research works adopting the above hybridization methods are discussed for better understanding. Eutectic salt hydrates PCM were melted to liquid phase followed by dispersion of $\alpha\text{-Al}_2\text{O}_3$ nanoparticles to enhance the performance of PCM without phase segregation. The mixture was ultrasonicated at 120W for 30 min to ensure proper dispersion [111]. Calcium chloride hexahydrate PCM and boron nitrate nanoparticles were mixed using the sonication microprobe technique at 40000 Hz for 15 min and was cooled in a water bath to avoid rise in temperature [112]. Nano-activated carbon and sodium fluoride nanoparticles were dispersed into eutectic salt hydrate PCM using a magnetic stirring technique to suppress the degree of

supercooling [113]. Calcium chloride hexahydrate was mixed with graphene oxide (GO) using a mechanical agitator for 30 min in a water bath at 40°C temperature [114]. Hand-mix techniques are the commonly adopted dry blending method, which is used for mixing of PCM and nanoparticles for building application, where the quantity is high.

Figure 11 comprises the spectroscopic evidence for hybridization of nanoparticles with salt hydrate PCM. Figure 11(a) shows the dispersion of EG and TiO₂ on calcium chloride hexahydrate PCM salt [115]. EG are the worms-like structure comprising of graphite flakes with a smooth surface. Thin titanium oxide layers are attached to the graphite flakes of modified EG. In Figure 11(b), the needle-like structure indicates DSP salt hydrate, which is filled into the inner channels and interlayers of EG; it also ensures the uniform dispersion of CNF within DSP salt hydrate [116]. This clearly indicates the successful dispersion of CNF and EG with DSP salt hydrate for improving the thermal conductivity of the composite PCM. Figure 11(c–e) shows the spectroscopic evidence and structure of EG, SAT PCM with EG and SAT with EG and CuS nanoparticle, respectively [117]. EG has a wormlike structure with a honeycomb network in which the SAT PCM salt hydrates are impregnated within the inner space of EG. As well, the CuS nanoparticles are attached to the fracture surface of SAT + EG structure in small dense particles. The CuS nanoparticles contribute towards enhanced thermal property and the EG structure offers phase stabilization and initiation nucleation. The micrograph in Figure 11(f) shows the dispersion of graphite and CNF nanoparticles using the freezing drying technique [97]. It depicts strong fibril–fibril interaction of CNF to enhance the network between graphite and salt hydrate. The spectroscopic image provides evidence of high porosity and oriented walls for the formation of foam stable PCM. Dispersed CNF promotes a homogeneous distribution of the salt crystals onto the graphite lamellae. The resulting foam is thus characterized by a porous structure with composite cell walls where the salt crystals are embedded in a G/CNF matrix. The morphological TEM image of CNF/GNP composite is shown in Figure 11(g) [89]. The CNF nanoparticles exhibit morphology of a high aspect ratio and entangled network structure, whereas GNP shows several layers of graphene that are stacked together due to the effect of agglomeration, this could be rectified with hybridization of a lower percentage of GNP with CNF. The SEM image in 11 h shows the CNF nanoparticles that form a bridge between SAT salt hydrates and GNP for uniform dispersion.

Metal additives are continuous and possess high strength structure, thereby improving the form stable of salt hydrates. On the contrary, salt-hydrated PCM undergo thickening treatment prior to inclusion of metal foam in order to ensure (a) stronger binding

potential with metal (b) to avoid loss of latent heat. The need for a thickening agent is owing to the fact that pores of metal foam and metal additives are larger. Comparative analysis on dispersion techniques of fullerenes, CNT/CNF, graphene and graphite with salt hydrate PCM is critically discussed and analysed. It is noted that very minimal work on the dispersion of fullerene with salt hydrate PCM is carried out due to its low thermal conductivity. However, CNT/CNF, graphene and graphite nano additives highly contribute towards enhancing thermal conductivity of salt hydrate due to their unique property and structure. Very keen identification ensures that dispersion of GO with salt hydrate is very poor and gets agglomerated. The cost of nanoparticles varies with respect to the surface area, size and dimension. Price values of different nanoparticles as supplied by one of the leading CNT manufacturers Timesnano will provide a better understanding regarding the importance of economic analysis. Pure SWCNT cost 200 \$ g⁻¹, DWCNT with 2–4 nm cost 42 \$ g⁻¹ and MWCNT cost 8 \$ g⁻¹. Similarly, pure graphene cost 200 \$ g⁻¹, GO powder cost \$130 g⁻¹ and graphite oxide powder costs \$5 g⁻¹ [118]. As per the cost details from Sigma Aldrich, polymer nano additive-like conducting polymer polyaniline cost 45 \$ g⁻¹ and polypyrrole cost 66 \$ g⁻¹ [119]. Metal nano additive aluminium oxide and copper oxide cost around 10 \$ g⁻¹, whereas metal foam-like the copper foam of size (6 mm * 200 mm * 300 mm) cost 85 \$ piece⁻¹ and nickel foam of size (200 mm * 300 mm * 0.3 mm) cost 45 \$ piece⁻¹ as per the details provided from Xiamen Tmax Battery Equipment limited. On the contrary, the cost of salt hydrate is very cheap and the commonly available CaCl₂·6H₂O salt cost 0.2 \$ kg⁻¹. Based on the cost details provided above, it can be inferred that even 0.1% of graphene inclusion with 1 kg of CaCl₂·6H₂O will incur 200 \$ excess, which is really costly. Thus high cost of nano additives drastically retracts its application on a commercial level. In addition, they lack a fully developed *in situ* technique for synthesis and hybridization of nano additive for mass application. Research works on polymer additive salt hydrate are on the limited number as polymer materials are mostly used as encapsulation shells. Additives like polyurethane foam mitigate phase separation and crystallization due to the presence of micro pores. In common, polymer additives act as supporting skeleton for providing enhanced thermal insulation.

Advances in enhancement and deterioration of thermal properties

The thermal property of inorganic salt hydrate PCM decides its application for storing thermal energy in various sectors. TES using salt hydrate are widely attracted among researchers due to their apparent benefits of energy storage density, cost, non-

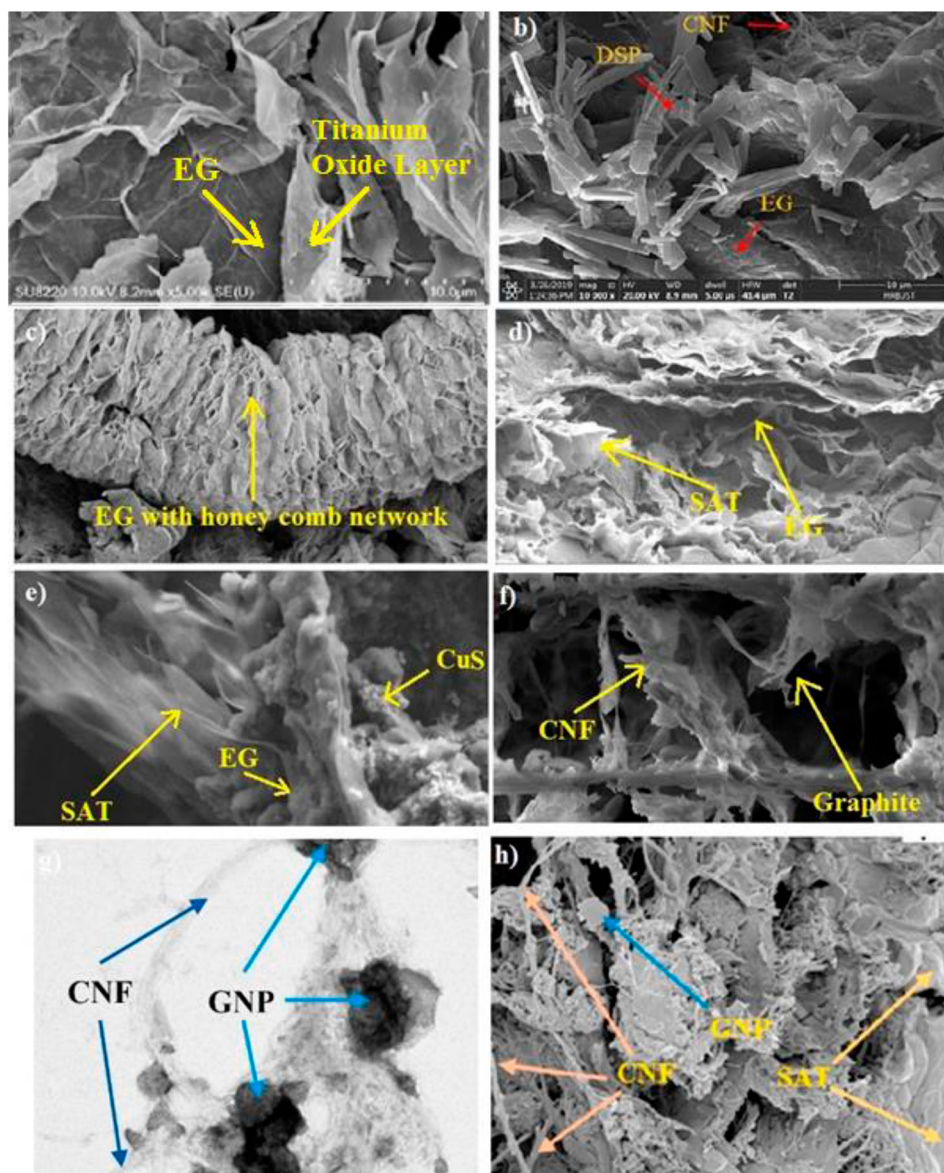


Figure 11. Spectroscopic evidence for ensuring hybridization. (a) EG + TiO₂ with CaCl₂·6H₂O salt hydrate [115] (with permission from Elsevier), (b) CNF + EG with DSP [116] (with permission from IOP), (c) EG structure, (d) EG + SAT salt hydrate, (e) EG + CuS SAT salt hydrate [117] (with permission from Elsevier), (f) graphite + CNF with salt hydrate [97] (with permission from Elsevier), (g) TEM image of CNF + GNP nanoparticle and (h) SEM Image of CNF + GNP with SAT salt hydrate [89] (with permission from Elsevier).

inflammable and non-toxic in nature. Nevertheless, salt hydrate has to overcome major issues like supercooling, low thermal conductivity, phase separation and corrosive nature to propel their application in commercial products. Techniques like doping of nano additive, impregnation into porous medium, encapsulation using polymer materials are adopted to overcome the drawbacks. This section critically reviews the advancement and deterioration of thermal properties of salt hydrate on hybridizing with nano additives.

Thermal conductivity

Mechanism and factors influencing thermal conductivity

Thermal conductivity is a parameter that replicates the heat transfer rate of PCM, and it indicates the ability of

a material to transmit heat energy. In solid materials, thermal conductivity occurs due to atomic vibration and due to conduction by free electrons. When thermal energy comes in contact with material, the surface atom of the material starts vibrating and transmits the energy to the adjacent atom and the wave proceeds and the energy diffuses till it remains in contact with atom. In ceramic and polymer materials, thermal conductivity due to phonon are predominant, whereas in pure metals, heat transfer by free-electron are higher [120]. Inorganic salt hydrate have their electrons bound to the ionic crystal lattice and the very few free electrons do not act effectively as heat conduction carriers, thus inorganic PCM depends on the vibration of crystal lattice for heat transfer. Figure 12(a) illustrates the mechanism of thermal conductivity propagation by lattice vibration and transfer of phonon. The SEM image in

Figure 12(b) shows the low thermal conductivity region of disodium phosphate salt hydrate PCM and Figure 12(c) depicts the CNF network for thermal conductivity improvement. Thermal conductivity by particle collision is well depicted in Figure 12(d). The major influencing factors of thermal conductivity are (a) nature of material, (b) size and fraction of material, (c) morphology, (d) heat capacity, (e) phonon group velocity and (f) phonon mean free path [121]. Plastic does not conduct as its electrons are not free and are bound to the parent atom. The size of the materials decides the time of travel, the thicker the material more is travel time taken. Thermal conductivity of salt hydrate with CNT and GP increase for two reasons (a) presence of high thermal conductive carbon nano additive where graphite has a thermal conductivity of $200\text{--}250\text{ W m}^{-1}\cdot\text{K}^{-1}$ and CNT has a thermal conductivity of $2500\text{--}3000\text{ W m}^{-1}\cdot\text{K}^{-1}$ (b) even dispersion of nano additives into the salt hydrate.

Effect on thermal conductivity with composite additive

High thermal conductivity shows the potential to enhance the heat transfer rate and consistent temperature distribution in thermal energy storage units. The thermal conductivity of salt hydrate PCM is extremely improved with the hybridization of nano additives. The thermal conductivity of salt hydrate not only depends on the property and concentration of doping additives but also on the morphological structure of the doped additive. Irrespective of the matrix of additives dope, the microstructure of salt hydrate is affected and the thermophysical properties are varied. Even with an increase in the number of thermal conduction channels and reduction in phonon scattering, the thermal conductivity of salt hydrate is improved.

Metal foam, due to their larger surface area per unit volume, their skeleton structure and high porosity voids, make themselves competitive with carbon additives to improve the thermal conductivity of salt hydrate on hybridization. Presently aluminium foam, copper foam, titanium oxide foam and nickel foam are hybridized with PCM to improve thermal properties. Copper foam with their individual high thermal conductivity of $410\text{ W m}^{-1}\cdot\text{K}^{-1}$ and owing to their cellular structure, they are widely used [122]. Open-hole aluminium foam, due to their low density, individual high thermal conductivity of $167\text{ W m}^{-1}\cdot\text{K}^{-1}$ and large surface area they are also used to improve thermal conductivity [123]. Mustaffar et al. [74] ensured irregular aluminium foam to improve thermal conductivity of salt hydrate by 24 times owing to better heat transfer characteristic. Nickel foam exhibit three-dimensional network structure, high porosity and high surface area making themselves suitable to advance the thermal conductivity of salt hydrate [124]. Titanium dioxide metal foam is environmentally friendly, cheaper and

fireproof material with high thermal conductivity. Among carbon nano additives, CNF/CNT and graphene are preferable for obtaining high improvements in thermal conductivity, though there are economically costlier. As CNF have large axial dimensions, it can spread comfortably within the salt hydrate medium; as well, the smaller diameter of CNF ensures easy preparation from EG. In addition, rather than single EG support, a combination of CNF/EG additives is preferred to enhance the thermal energy transfer at a higher rate. Similarly, DSP [116] was doped with CNF/EG, where the characterization results showed an increase in thermal conductivity of pure salt hydrate by 3.69 times, which is achieved due to the presence of CNF/EG in the low thermal conductivity spots of salt hydrate. Though supporting materials with porous supports improves the thermal stability and thermal conductivity of salt hydrates, few porous structures of EG exhibits more number of low thermal conductivity spots, and this major issues are of high research spot to optimize the porous structure to improve the thermal conductivity. An increase in spacing between layers of graphite oxide sheets also increases the thermal conductivity of hydrate salts and eutectic salts [125].

In addition to Table 2, Figure 13 shows the variation in thermal property of salt hydrate PCM due to loading of nano additives. The variation taken for the comparison are the best results obtained; however, their no constant trend in the increase or decrease of property with respect to loading concentration of nano additives in salt hydrate PCM. Thermal conductivity of salt hydrate PCM was observed to increase by almost 2000% on the dispersion of EG; nevertheless, research work by certain authors shows a decrease in thermal conductivity of salt hydrate PCM with a dispersion of EP. It is clearly observed that the maximum thermal conductivity for nano additive salt hydrate PCM obtained is $10.93\text{ W m}^{-1}\cdot\text{K}^{-1}$, however experimental research work by Wu et al. [126] gives us an understanding of the technique to load 40% of additives with PCM to obtain thermal conductivity in the range of $10\text{--}35\text{ W m}^{-1}\cdot\text{K}^{-1}$. Further research focusing to improve the thermal conductivity of PCM to a range of $35\text{ W m}^{-1}\cdot\text{K}^{-1}$ will be a highly novel work. Likewise, inclusion of nanoparticles like carbon additives and metal foams, the thermal conductivity of salt hydrate can also be improved by encapsulation and by using fins made of non-corrosive metals.

Latent heat and heat capacity

Mechanism and factors influencing latent heat and specific heat

Latent heat is the amount of heat that can be stored within a defined mass of material while melting and can be liberated during solidification. Latent heat

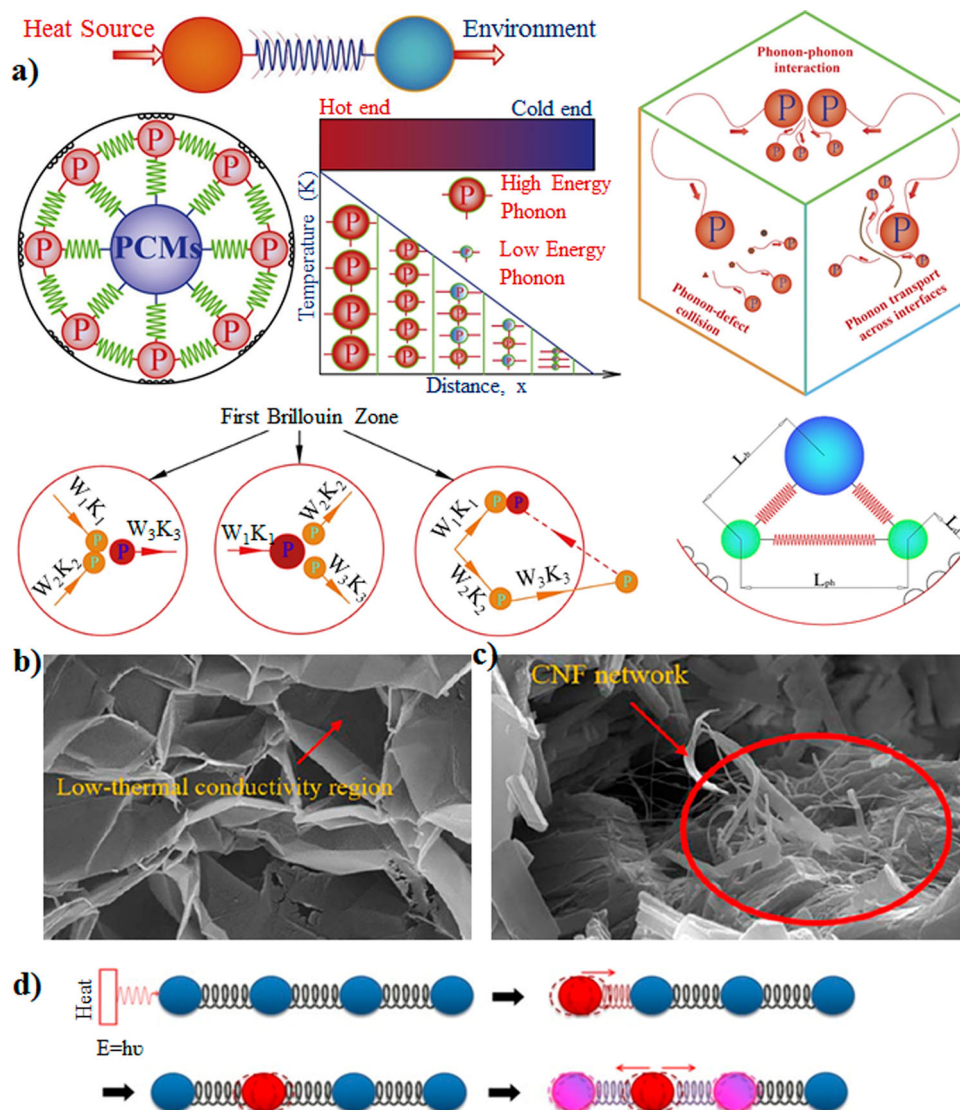


Figure 12. (a) Thermal conductivity mechanism by lattice vibration and phonon transmission [121] (with permission from Elsevier), (b) SEM image of low thermal conductivity region, (c) CNF network that improves the thermal property [116] (with permission from IOP) and (d) Thermal conductivity by particle collision [120] (with permission from Elsevier).

stores energy within the particles of the material, phase transfer occurs during the transfer of heat and potential energy within a substance, resulting in storage of thermal energy without significant change in temperature [127]. Latent heat indicates the amount of thermal energy that can be stored within one kilogram of the substance. The higher the latent heat value of the substance, the lesser is the mass of the substance required. Less mass occupies less space, which reduces the storage system cost. So salt hydrate PCM with higher latent heat is considered to be very effective for storing thermal energy. In addition, specific heat indicates the amount of thermal energy required to raise the temperature of 1 g of substance by 1°C. Latent heat and specific heat capacity are directly proportional. For TES system, salt hydrate PCM with high latent heat and high specific heat capacity are preferred, as they can store more energy. Factors affecting latent heat are (a) mass of the substance and (b) material morphology.

Effect on latent heat and heat capacity with composite additive

Latent heat of fusion for inorganic salt hydrate normally varies between 125 and 280 J g⁻¹, for low- and medium-temperature application and varies between 250 and 450 J g⁻¹ for high-temperature application. And it is quite obvious that latent heat values of inorganic salt hydrates are higher than most organic PCM [128]. Greater the latent heat, the greater is the TES potential. The inclusion of nanoparticles with salt hydrate to improve the thermal property tends to make a negative impact on latent heat and deteriorates due to changes in size and surface characteristics [129]. It is also noted that supporting material with narrow pores affects the crystalline behaviour of the material and will result in the decline of latent heat as the material will not be able to reach the lowest energy state.

By incorporating graphite, MWCNT and nanocellulose with Mg(NO₃)₂·6H₂O, the composite PCM

Table 2. Summary on enhancement and deterioration of thermal property on dispersion of nano additives with salt hydrate.

Salt hydrate	PCM ratio (wt-%)	Without additives				Nano additives	With additives				Change in %				Ref.
		T_{mi} (°C)	H_{mi} (J kg ⁻¹)	T_{si} (°C)	k_i (W m ⁻¹ ·K ⁻¹)		T_{mf} (°C)	H_{mf} (J kg ⁻¹)	T_{sf} (°C)	k_f (W m ⁻¹ ·K ⁻¹)	ΔT_m (°C)	ΔH_m (J kg ⁻¹)	ΔT_s (°C)	Δk (W m ⁻¹ ·K ⁻¹)	
CH ₃ COONa · 3H ₂ O	80.6	58.3	286.3	32	—	EG, Na ₂ HPO ₄	57.8	218.6	7.1	—	−0.9	−23.6	−77.8	—	[117]
CH ₃ COONa · 3H ₂ O	72.5					EG, Na ₂ HPO ₄ , CuS	56.9	194.5	7.8	—	−2.4	−32.1	−75.6	—	
	92.9	55.1	267.2	0.2	0.71	Hp-GNP750, Na ₂ HPO ₄	55.1	248.3	0.1	1.52	0.0	−7.1	−50.0	114.1	[86]
	89.1					Hp-GNP500, Na ₂ HPO ₄	55.2	238.2	0.2	1.69	0.2	−10.9	0.0	138.0	
	86.4					Hp-GNP300, Na ₂ HPO ₄	54.9	230.8	0.3	1.93	−0.4	−13.6	50.0	171.8	
CH ₃ COONa · 3H ₂ O	96.25	58.53	242.4	0.5	0.777	Nano-Cu	57.88	231.4	3.527	0.936	−1.1	−4.5	605.4	20.5	[143]
98%SAT and 02% Xanthan gum	77.2	58.8	255.5	2.6	0.76	Copper foam	58.8	197.2	2.1	2.10	−1.1	−22.8	−19.2	176.3	[144]
CH ₃ COONa · 3H ₂ O	95	58	—	28.3	—	Al ₂ O ₃ /CMC	52.1	—	14.2	—	0.0	—	−49.8	—	[145]
92%SAT and 08%KCl	95	50.42	236.17	7.3	—	Al ₂ O ₃ /CMC	49.79	241.2	0.1	—	−10.2	2.1	−98.6	—	
CH ₃ COONa · 3H ₂ O	96.5	56.75	271.09	6.63	0.59	DSP/CMC	60.29	250.5	4.34	—	−1.2	−7.6	−34.5	—	[68]
	95.5	56.75	271.09	6.63	0.59	Cu foam, DSP/CMC	—	—	3.54	6.8	—	—	−46.6	1052.5	
CH ₃ COONa · 3H ₂ O	95.95	58.74	261.0	8.1	0.868	α-Fe ₂ O ₃	58.66	251.2	0	1.063	−0.1	−3.8	−100	22.5	[146]
CH ₃ COONa · 3H ₂ O	70	48.45	240.4	5.05	0.672	(SiO ₂), urea	35.75	151.6	1.14	0.310	−26.2	−36.9	−96.1	−53.9	[65]
CH ₃ COONa · 3H ₂ O	85.4	58	264	40	—	DSP, CMC, EG	58.54	247	4	—	0.9	−6.4	−90.0	—	[147]
CH ₃ COONa · 3H ₂ O	92.4	58.4	249.1	0.8	0.61	DSP, CMC, EG	58.	210.7	2	1.37	−0.7	−15.4	150.0	124.6	[148]
CH ₃ COONa · 3H ₂ O	95.75	59	258	29	0.7	Nano whiskers, CMC	60	241	0.9	0.74	1.7	−6.6	−96.9	5.7	[149]
	89.75					Nano whiskers, CMC, hBN	60	210	0.9	1.2	1.7	−18.6	−96.9	71.4	
75% SAT — 25% FA	97	40.88	233.9	2.15	0.79	EG, DSP	40.3	226.3	1.57	2.46	−1.4	−3.2	−27.0	211.4	[150]
	95						40.7	215.8	0.63	2.5	−0.5	−7.7	−70.7	216.5	
	92						37.5	187.6	0.81	3.11	−8.2	−19.8	−62.3	293.7	
	90						37.7	177	1.85	3.47	−7.8	−24.3	−14.0	339.2	
88%CH ₃ COONa · 3H ₂ O — 12%Glycine	96	48.62	258.5	3.62	4.151	EG, borax	48.0	235.0	2.08	ND	−1.3	−9.1	−42.5	—	[151]
	92						47.2	224.9	2.67	ND	−2.9	−13.0	−26.2	—	
	88						47.1	214.7	1.49	ND	−3.0	−16.9	−58.8	—	
	84						46.6	207.5	3.31	ND	−4.1	−19.7	−8.6	—	
MgCl ₂ · 6H ₂ O	80.1	116.69	144.2	11.2	0.465	Modified EG, TX-100	116.24	116.7	1.3	3.588	−0.4	−19.1	−88.4	671.6	[92]
MgCl ₂ · 6H ₂ O	65	115.4	137.5	—	—	Carbon aerosol	115.6	122.6	—	—	0.2	−10.8	—	—	[152]
Mg(NO ₃) ₂ · 6H ₂ O	99.5	—	—	—	0.58	Nanographite	—	—	—	2.165	—	—	—	273.3	[130]
Na ₂ HPO ₄ · 12H ₂ O	98	36.3	207.9	14.5	—	CNT	33.44	174.8	4.0	—	−7.9	−15.9	−72.4	—	[153]
						EG	33.89	192.0	11.0	—	−6.6	−7.6	−24.1	—	
						GNS	34.73	202.9	0.4	—	−4.3	−2.4	−97.2	—	
NaSO ₄ · 10H ₂ O	87	33.4	226.90	27.66	—	EG, Na ₂ HPO ₄	32.0	172.3	14.94	3.643	−4.0	−24.1	−46.0	—	[154]
NaSO ₄ · 10H ₂ O	83.7	32.24	139.9	27.31	0.544	EG, borax	31.0	114.0	8.03	1.96	−3.7	−18.5	−70.6	260.3	[96]
Na ₂ SO ₄ · 10H ₂ O	67	32.5	250.2	10	—	Sodium alginate/PAAM	38.1	202.9	2.9	—	17.2	−18.9	−71.0	—	[155]
Na ₂ SO ₄ · 10H ₂ O	93	34.7	235.7	20	0.5723	MWCNT, PAAS	38.5	187.2	8.1	1.3831	11.0	−20.6	−59.5	141.7	[156]
KAl(SO ₄) ₂ · 12H ₂ O	90	91.11	250.8	—	0.497	EG	87.9	214.9	—	5.875	−3.5	−14.3	—	1082.1	[133]
KAl(SO ₄) ₂ · 12H ₂ O	91.4	76.54	136.3	2.238	0.4991	EG, Sod	75.5	123.5	1.866	10.93	−1.4	−9.4	−16.6	2089.9	[157]
Al ₂ (SO ₄) ₃ · 18H ₂ O	86.5	—	—	26.7	—	TiO ₂	—	—	10	—	—	—	−62.5	—	[158]
Ba(OH) ₂ · 8H ₂ O	85.8	78.3	256.4	13	0.4972	Modified EG	77.8	238.4	2.4	3.58	−0.6	−7.0	−81.5	620.0	[159]
Na ₂ HPO ₄ · 12H ₂ O	80.2	35.4	260.5	—	0.674	EG-CNF	33.7	208.9	—	1.747	−4.8	−19.8	—	159.2	[116]
80% CaCl ₂ · 6H ₂ O—15% urea—5% Ethanol	85	11.54	123.1	0.94	0.4544	EG-50, SrCl ₂ · 6H ₂ O	10.7	106.1	0.19	3.283	−7.3	−13.8	−79.8	622.5	[93]
						EG-80, SrCl ₂ · 6H ₂ O	10.5	102.5	0.23	2.806	−8.8	−16.7	−75.5	517.5	
						EG-100, SrCl ₂ · 6H ₂ O	10.1	93.03	0.44	2.616	−11.8	−24.4	−53.2	475.7	
						SrCl ₂ · 6H ₂ O	28.65	151.3	—	—	0.9	−0.7	—	—	[160]
CaCl ₂ · 6H ₂ O	90	28.39	152.4	—	0.6	EG	27.11	118.7	—	3.2	−4.5	−22.1	—	433.3	

CaCl ₂ · 6H ₂ O	82.6	11.54	123.1	0.95	0.4544	EG, TiO ₂	10.7	88.39	0.18	8.83	−7.5	−28.2	−81.1	1843.2	[115]
CaCl ₂ · 6H ₂ O	55	28.11	159.85	–	0.6	EP, SrCl ₂ · 6H ₂ O	27.3	87.44	–	0.17	−2.6	−45.3	–	−71.7	[161]
CaCl ₂ · 6H ₂ O	–	28.96	151.6	20	0.529	EG	27.74	121.8	7	2.065	−4.2	−19.7	−65.0	290.4	[162]
	–	28.96	151.6	20	0.529	EG, photo-curing polymer	28.53	113.2	7	1.536	−1.5	−25.3	−65.0	190.4	
CaCl ₂ · 6H ₂ O	99.5		178	5.6	0.341	γ-Al ₂ O ₃	–	176	0.2	0.459	–	−1.1	−96.4	34.6	[129]
	99						–	171	0.3	0.672	–	−3.9	−94.6	97.1	
	98.5						–	163	0.4	0.976	–	−8.4	−92.9	186.2	
	98						–	159	0.8	1.373	–	−10.7	−85.7	302.6	
SAT-FA	55	42.4	274.2.1	34.5	0.6348	EP, Na ₂ P ₂ O ₇ · 10H ₂ O	40.5	148.3	0.4	0.0978	−4.5	–	−98.8	−84.6	[163]
CH ₃ COONa · 3H ₂ O	85.7	58.1	277.7	–	0.6	EV, Na ₂ B ₄ O ₇ · 10H ₂ O	57.6	238.8	–	0.8	−0.9	−14.0	–	33.3	[164]
CaCl ₂ · 6H ₂ O	58.3	28.4	185.6	–	–	Diatomite	28.9	108.2	–	–	1.8	−41.7	–	–	[165]
Mg(NO ₃) ₂ · 6H ₂ O	50	89.91	148.34	16.08	0.73	Diatomite	88.9	69.39	0.3	0.9194	−1.1	−53.2	−98.1	25.9	[166]
BaCl ₂ · 2H ₂ O	99	–	269	6.5	0.6146	MWCNT	–	236	–	0.6279	–	−12.3	–	6	[167]
	99					MgO		247.5	–	0.7913	–	−8	–	17	
Ba(OH) ₂ · 8H ₂ O	91	78.2	270.8	–	1.163	EG	77.2	241.6	–	4.672	−1.3	−10.8	–	301.7	[168]
	97						77.8	245.7	–	3.662	−0.5	−9.3	–	214.9	
	95						77.8	253.2	–	3.055	−0.5	−6.5	–	162.7	
	97						78.2	261	–	2.221	0.0	−3.6	–	91.0	
	99						77.9	267.7	–	1.709	−0.4	−1.1	–	46.9	

Note: T_m – phase change temperature, H_m – latent heat, T_s – degree of supercooling, k – thermal conductivity, [subscript (i – initial, f – final)].

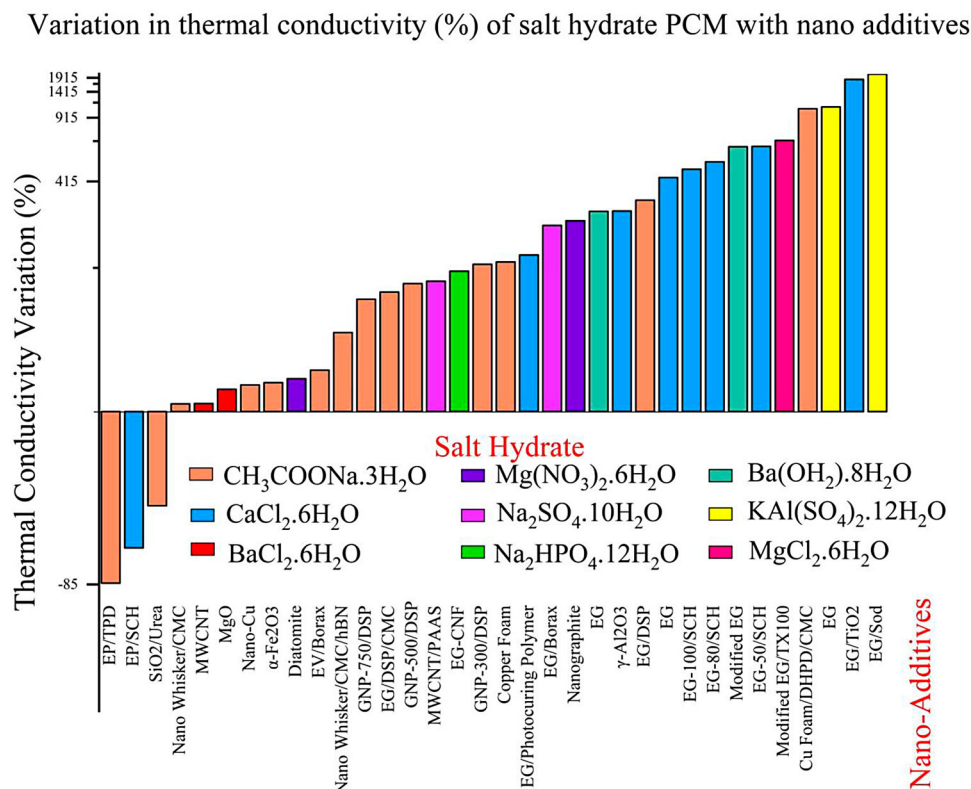


Figure 13. Variation in thermal conductivity of salt hydrate due to loading of nano additives.

exhibited quick heating on exposure with sunlight; however, the latent heat of composite PCM declined with an increase in the concentration of nano particle [130]. In another investigation, $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ salt hydrate is confined within solid nano bowls of SiO_2 matrix. The encapsulation technique is carried out to address the issue of phase segregation and in addition, it is observed that the encapsulation technique effectively suppresses the degree of supercooling, which is an effect of mesoporous confinement. Here the weight percentage of salt hydrate with SiO_2 is 58%. And from the DSC results it is found that the enthalpy of $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ is 254 J g^{-1} and of $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}@\text{SiO}_2$ is 180.7 J g^{-1} , which clearly indicates an increase in 70% of latent heat of salt hydrate though the composite consists of only 58% of pure salt hydrate [131]. With metal additives like aluminium oxide, latent heat of $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ decreases due to inhibition of salt hydrate crystallization [129], whereas with GO, the latent heat of $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ improved by 13% [114]. Figure 14 shows variation in latent heat of salt hydrate on doping nano additives. It is clear that dispersion of nano additives with salt hydrate PCM resulted in the drop of latent heat of salt heat by a maximum of 54%; however, the majority of nano additives reduced the enthalpy change by 10–20% which was under permissible value. Salt hydrate PCM offers 17.5% higher latent heat value compared to organic PCM [132]; hence any drop in enthalpy change below 20% is expected to run out of the

scope for which the particular salt hydrate PCM was selected. Close observation ensures drop-in latent heat of salt hydrate with organic nanoparticles, still the drop was marginally low and does not affect the TES storage performance of salt hydrate as the nano additives make remarkable improvement with thermal conductivity and also suppress the effect of supercooling and phase separation.

Melting point

Mechanism and factors influencing melting point

The selection of inorganic salt hydrate PCM for any application is based on their melting point. The melting temperature of salt hydrate is recommended to be below the temperature of the heat source and above the temperature of ambient condition for TES. Melting temperature or phase change temperature indicates the critical point during phase transfer from solid to liquid and vice versa. For refrigeration application PCM with 21°C , for air conditioning application $21\text{--}28^\circ\text{C}$, for energy saving in buildings $21\text{--}60^\circ\text{C}$ and for water heating application $29\text{--}80^\circ\text{C}$ are preferred. The higher melting point of salt hydrate contribute towards a longer melting duration and provides better thermal cycle before complete melting of PCM. The major factors influencing melting point are (a) intermolecular forces (b) molecular bonding and (c) shape and size of molecule.

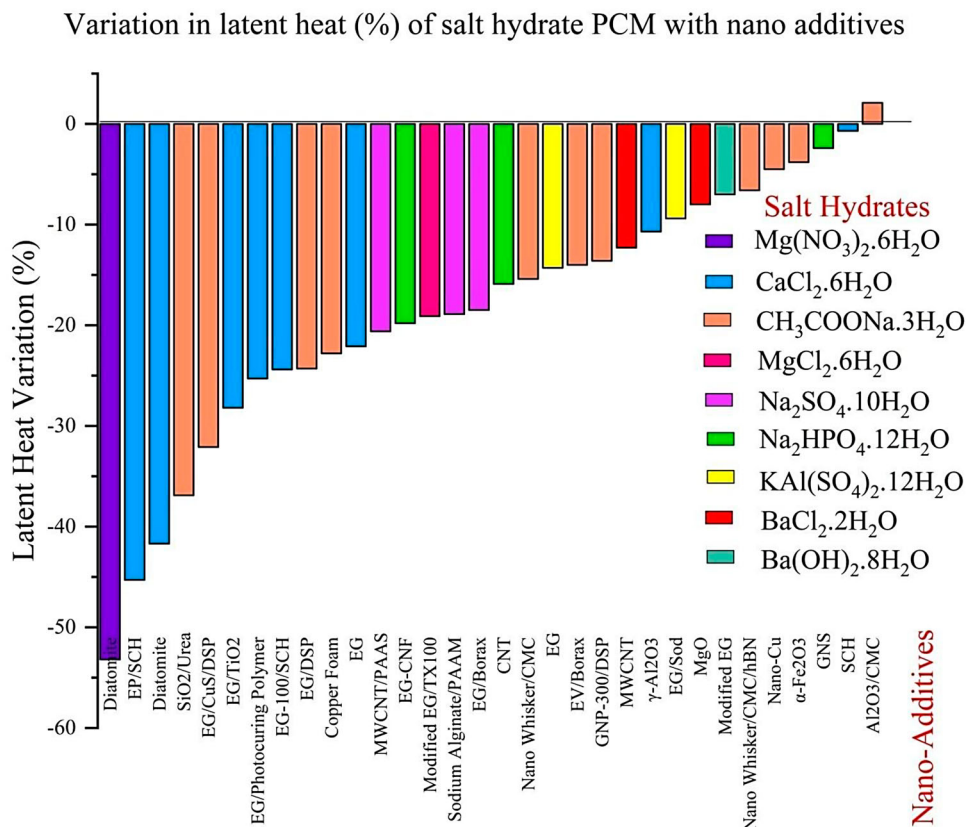


Figure 14. Variation in latent heat of salt hydrate due to loading of nano additives.

Effect on melting point with composite additive

As the PTT of most inorganic salt hydrate ranges between 20°C and 140°C they are widely used in low- and medium-temperature TES system. Due to the variation of thermal conductivity and due to interaction in bonds between support and hydrate salt, there exist a small deviation in the melting point of pure salt and the composite salt hydrate. The inclusion of nano additives into salt hydrate modifies the force of attraction between the molecules and ions of salt hydrate and varies the melting point slightly. It has been noted that additives with high thermal conductivity have a small decreasing tendency of melting point and vice versa. Zhang et al. [133] mixed EG with aluminium potassium sulphate dodecahydrate K₂Al(SO₄)₂·12H₂O and observed melting point of pure alum dive to 87.92°C from 91.11°C, from the DSC curve results, it was ensured that the EG provides heat conduction path in the alum and it decreases the melting point of salt hydrate. Similar results with a decrease in melting point on the inclusion of high thermal conductivity nano additive are seen in Table 2. As well, nano additives that express a strong affinity with salt hydrates exhibit a minor increasing tendency of melting point. In addition, encapsulation of salt hydrate within a narrow space of supporting additive the melting point of PCM greatly varies. Encapsulation technique and encapsulation material used by Liu et al. [134] improved the PTT of composite PCM by 13.7°C, however, encapsulation research

work of Fu et al. [65] depicts a drop in PTT by 14.09°C. The justification provided for the variation in melting point of salt hydrate due to confinement was the presence of irregular pore configuration of the supporting material and the resistance provided by the capsule. The most attractive results obtained were stability in melting point of CaCl₂·6H₂O after 1000 thermal cycles without the inclusion of nano additives and thickening agent [135]. Figure 15 shows the variation in melting point of salt hydrates due to doping of nano additives, and it clearly depicts variation in melting point by -20% to +20%.

Degree of supercooling

Mechanism and factors influencing super cooling

Supercooling is a phenomenon where the PCM is noted to be in a liquid state even below the fusion temperature. Supercooling effects delay the recovery of the heat energy stored within salt hydrate. Almost all inorganic salt hydrate PCM exhibit supercooling effect than organic PCM. On a note nucleation agents are incorporated with PCM to trigger the crystallization process in salt hydrate and reduce the supercooling effect. In PCM research, better understanding of melting and crystallization processes are important. Crystallization takes place by the following three steps (a) induction duration (b) growth phase of crystal and (c) regrowth of crystal. In the induction phase, the nuclei develop and reach a sufficient size; then

nucleation initiates followed by diffusion to the nucleus and gets adsorbed with the surface. Supercooling or sub cooling is an intrinsic property of material; in addition inherent and non-pertinent factors also affect the rate of supercooling. The major factors influencing melting point are (a) cooling rate (b) sample size (c) encapsulation medium (d) chemical purity (e) vibration (f) morphology of material and (g) crystallization temperature. Figure 16 points up few key techniques commonly adopted for the elimination of supercooling in salt hydrate PCM.

Effect on degree of supercooling with composite additive

The technique to reduce supercooling includes both passive and dynamic methods. In passive methods, supercooling is suppressed by homogeneous and heterogeneous nucleation. In heterogeneous nucleation, the suspended particles requires a high melting temperature compared to operating temperature, whereas during homogeneous nucleation thickening agents and surfactants are added. The suspended particles have a propensity to precipitate with a number of thermal cycles, as they prevent precipitation and ensure homogenous distribution. Though nucleating agent reduces supercooling, the level of mitigation varies with respect to different nucleation agents. Figure 17 shows salt hydrate and their corresponding nucleating agents used by researchers. Numerous research works to suppress supercooling using nucleating agents and nano additives are summarized in Table 2. Keen observation ensures that the effect on the concentration of nucleating agents with salt hydrate to mitigate supercooling is nonlinear. Effective nucleating agent concentration is 1–3% for reducing supercooling. The dynamic method includes crystallization of salt hydrate PCM by induced methods like agitation, supply of shock waves, ultrasonic vibration and manual shaking. Passive methods with nucleation agents are very efficient for incorporating salt hydrate for TES. On treating $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$, $\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O}$ and $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ salt with diatomite, the degree of supercooling of the pure salt was greater than 10°C , 40°C and 10°C and declined to 5.4°C , 7.8°C and 5.4°C , respectively. The suppression in the degree of supercooling was owing to the fact that the surface of diatomite has better hydrophilicity and tough wettability with hydrate molecules and propelled the contact area between salt hydrate PCM and diatomite, thereby offering more nucleation spots [67]. Aluminium nitride nanoparticles as nucleation agent and CMC as a thickening agent, abolish the supercooling effect of pure $\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O}$ [136].

Supercooling or degree of supercooling is a negative effect that mainly occurs in inorganic salt hydrate PCM. Depending on the concentration of nanoparticles deteriorates the supercooling effect. It is also

observed that organic nanoparticles make a remarkable reduction in supercooling of salt hydrate compared to the inorganic nanoparticles. In addition to Table 2, Figure 18 shows the supercooling variation due to doping of nano additives. Complete suppression of supercooling effect was observed with inclusion of $\alpha\text{-Fe}_2\text{O}_3$; on the contrary; supercooling effect increased by 600% on the dispersion of nano-Cu with salt hydrate PCM as they are the non-nucleating agent.

Resistance to corrosion

Mechanism and factors influencing corrosion

Depending on salt anion, corrosion mechanisms in salts are basically divided into (a) oxyanionic-based salt and (b) halide-based salt [137]. Similar to aqueous corrosion, corrosion mechanism in both types of salt usually occurs based on acid/base model. Molten salt hydrate carries negatively charged H ions where in aqueous solutions carries positively charged H ions. Carbonate, nitrates and sulphate salts fall under oxyanionic salt, during heating and melting dissociates. The propagation of oxyanionic salt highly depends on environmental conditions. In the case of halide salts like fluoride and chloride, corrosion involves halogenating transitional metals and is less corrosive due to the low activity of fluorine ions. Impurities and moisture content contribute towards corrosion due to the presence of oxide in the salt. The corrosion issues associated with hydrated salt can be mitigated with addition of corrosion inhibitors [138,139]. Factors that predominantly influence corrosion mechanism are (a) anion and cation of salt (b) temperature (c) pressure (d) thermal cycling rate (e) impurity concentration (f) environmental condition (g) moisture (h) microstructure and (i) composition.

Effect on corrosion with composite additive

Salt hydrate faces a major concern on affinity to corrode metals that are in contact with and the corroded metal diffuse into storage media of salt hydrates and disrupts the thermal property of salt hydrate also weakens the product. Corrosion involves the oxidation of metals and reduction of protons or oxygen. Protecting metals from corrosion improves their application and economic benefits. Conventional coating using paints, polyvinyl butyral and zinc chromate are familiar techniques followed to protect the metal. The coating layer acts as a barrier between oxygen and water resists the ionic movement between the salt and metal interface [87]. In some cases, coatings and additive doped to resist corrosion in metals with salt hydrate contact tends to create degradation in melting temperature, latent heat and thermal conductivity. The coating layer reduces corrosion, on the contrary, they acts as an insulator and reduces the

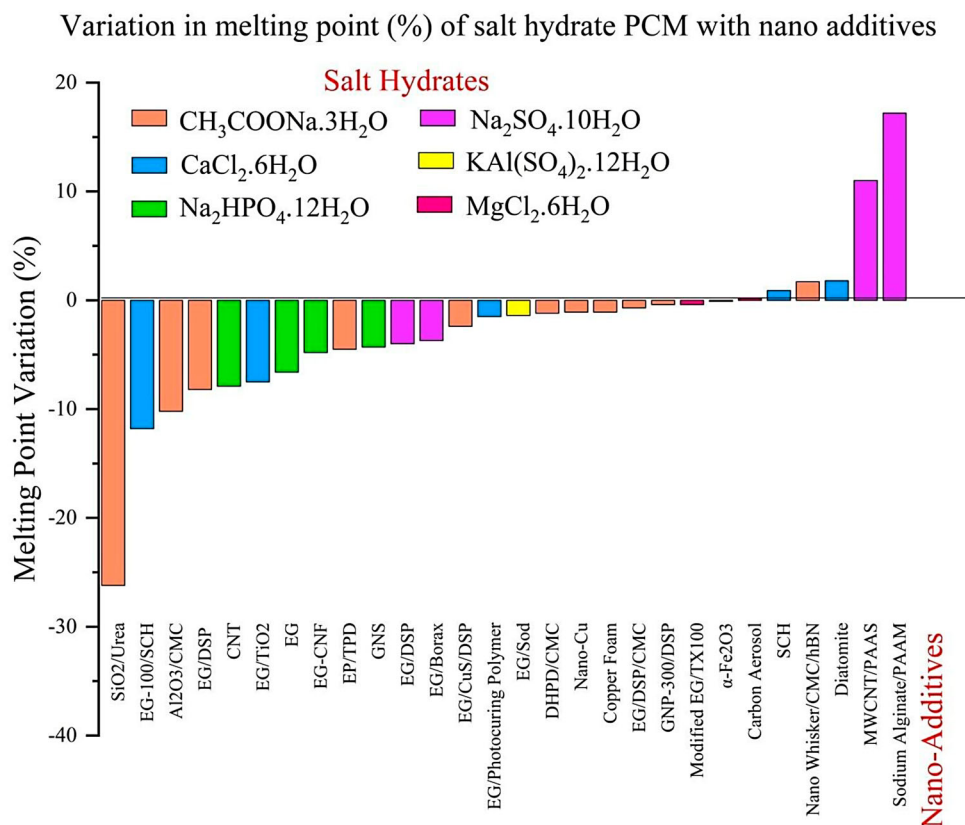


Figure 15. Variation in melting point of salt hydrate due to loading of nano additives.

tendency to transfer heat energy. As well, nucleating agents and additives incorporated into salt hydrate for enhancement of thermal properties have adverse effects on corrosion rate. $\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$ and $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ were mixed to form a composite PCM to overcome the corrosive nature of salt hydrate [72]. Corrosion inhibitor 2-(2-pentadecyl-4, 5-

dihydro-1H-imidazol-1-yl) ethanol was added up to an optimal limit of 0.5 mM. It's been observed that the corrosion inhibitor ethanol added to the salt hydrate reduced the mass transportation at the edge of Cu/PCM and reduced the corrosion rate of copper. Using $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ as salt hydrate PCM the corrosion nature of different aluminium alloys like

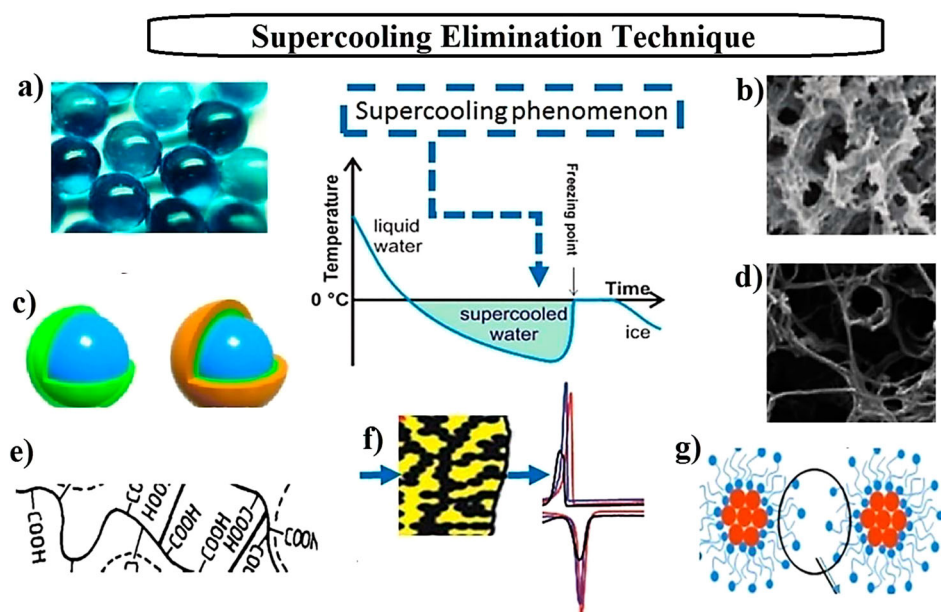


Figure 16. Key techniques adopted to eliminate supercooling in salt hydrate. (a) Microencapsulation, (b) mesoporous, (c) core shell and multi shell, (d) CNT networks, (e) cross-linking, (f) carbonized PCM with surfactants and (g) stable dispersion [18] (with permission from Elsevier).

Salt hydrate and their Nucleating Agent

Calcium Chloride Hexahydrate $(\text{CaCl}_2 \cdot 6\text{H}_2\text{O})$ • BaCO_3, • SrCO_3, • BaF_2, • SrF_2, • $\text{BaI}_2 \cdot 6\text{H}_2\text{O}$, • $\text{SrCl}_2 \cdot 6\text{H}_2\text{O}$, • $\text{SrBr}_2 \cdot 6\text{H}_2\text{O}$	Sodium dihydrogen phosphate dodecahydrate $(\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O})$ • Borax • Carbon • TiO_2 • Copper • Aluminium	Potassium Fluoride Tetrahydrate $(\text{KF} \cdot 4\text{H}_2\text{O})$ • Pumice stone	Lithium Chlorate Trihydrate $(\text{LiClO}_4 \cdot 3\text{H}_2\text{O})$ • KClO_4 • Na_2SiF_6 • K_2SiF_6 • BaSiF_6
Magnesium Chloride Hexahydrate $(\text{MgCl}_2 \cdot 6\text{H}_2\text{O})$ • Na_3AlF_6 • $\text{CaC}_2\text{O}_4 \cdot \text{H}_2\text{O}$ • $\text{Sr}(\text{OH})_2$ • SrCO_3 • $\text{Ba}(\text{OH})_2$ • $\text{Mg}(\text{OH})_2$ • $\text{Ca}(\text{OH})_2$ • BaO	Magnesium Nitrate Hexahydrate $(\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O})$ • $\text{MgSO}_4 \cdot 4\text{H}_2\text{O}$ • $\text{CuSO}_4 \cdot 3\text{H}_2\text{O}$ • $\text{NaBO}_2 \cdot 2\text{H}_2\text{O}$ • $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ • $\text{NiSO}_4 \cdot 4\text{H}_2\text{O}$ • SrCO_3 • MgO	Manganese(II) Nitrate Hexahydrate $(\text{Mn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O})$ • NaCl • NaNO_3 • NaSO_4 • Kerosene • Heptane • Carbon black • $\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$ • $\text{MnSO}_4 \cdot 6\text{H}_2\text{O}$	Sodium Acetate Trihydrate $(\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O})$ • Na_2SO_4 • SrSO_4 • Carbon • Potassium Sulfate • $\text{Na}_4\text{P}_2\text{O}_7 \cdot 10\text{H}_2\text{O}$ • Nano-Copper • $\text{Na}_2\text{SO}_4 + \text{AgNPs}$ • Borax
Potassium Fluoride Dihydrate $(\text{KF} \cdot 2\text{H}_2\text{O})$ • Al_2O_3	Sodium Sulphate Decahydrate $(\text{NaSO}_4 \cdot 10\text{H}_2\text{O})$ • Borax • Aerosol SiO_2	Sodium Thiosulfate Pentahydrate $(\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O})$ • K_2SO_4 • $\text{Na}_2\text{P}_2\text{O}_7 \cdot 10\text{H}_2\text{O}$	Zinc Nitrate Hexahydrate $(\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O})$ • ZnO • $\text{Zn}(\text{OH})_2$

Figure 17. Salt hydrates and the nucleating agents used for reducing supercooling.

Al2024, Al3003, Al6063 and Al1050 were studied [140]. Among the four salts, Al 2024 results in the formation of $\text{NaAlCO}_3 \cdot (\text{OH})_2$ due to the presence of atmospheric air, thereby indicating its non-compatibility for better corrosion resistance. Corrosion studies on salt hydrates like $\text{LiNO}_3 \cdot 3\text{H}_2\text{O}$, $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ and $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ with additives like zinc hydroxyl nitrate, zinc oxide/zinc hydroxyl nitrate and strontium chloride, respectively, were carried out [141]. The salt hydrates were made in contact with stainless steel and aluminium for 6–18 weeks and their resistance property was studied on 6th week, 12th week and 18th week in detail. As aluminium metals undergo pitting with salts of chloride, aluminium metal sample with $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ and its additive produced bubbles and changed from white cloudy solution to dark bluish over a period of time, and corrosion rate was about $127 \text{ mg cm}^{-2} \cdot \text{yr}^{-1}$, whereas the same salt hydrate with its additive on stainless steel metal showed a corrosion rate $2.4 \text{ mg cm}^{-2} \cdot \text{yr}^{-1}$ highlighting its application due to corrosion resistance property.

Several research were carried out to study the corrosive nature of pure salt hydrate PCM without the inclusion of additives with metals like stainless steel, aluminium, copper and carbon steel a decade ago. Recent work focuses on corrosion study and characterization of thermal properties of salt hydrate with additives suppressing other major issues like

supercooling, phase separation and low thermal conductivity. However, the corrosion studies on salt hydrate with carbon nano additives and polymer additives are yet to explore.

Phase separation

Mechanism and factors influencing phase separation

Phase separation deteriorates the TES potential of salt hydrate PCM due to the separation of salt and water molecules. On the propagation of time and favourable conditions, the salt solution prepared by melting transforms back to pure salt. This occurs when salt and water molecules are uniformly diffused and can reestablish their molecular structure. Due to the density difference between salt and hydrate, the anhydrous salt separates from salt hydrate solution under gravity and settles down. This is the mechanism of phase separation occurring in salt hydrate PCM. The factors that influence phase separation are (a) solubility and (b) temperature of the salt hydrate solution.

Effect on phase separation with composite additive

Phase separation due to incongruent melting of salt hydrate has a negative impact on its application to store energy. The inclusion of thickening and binding

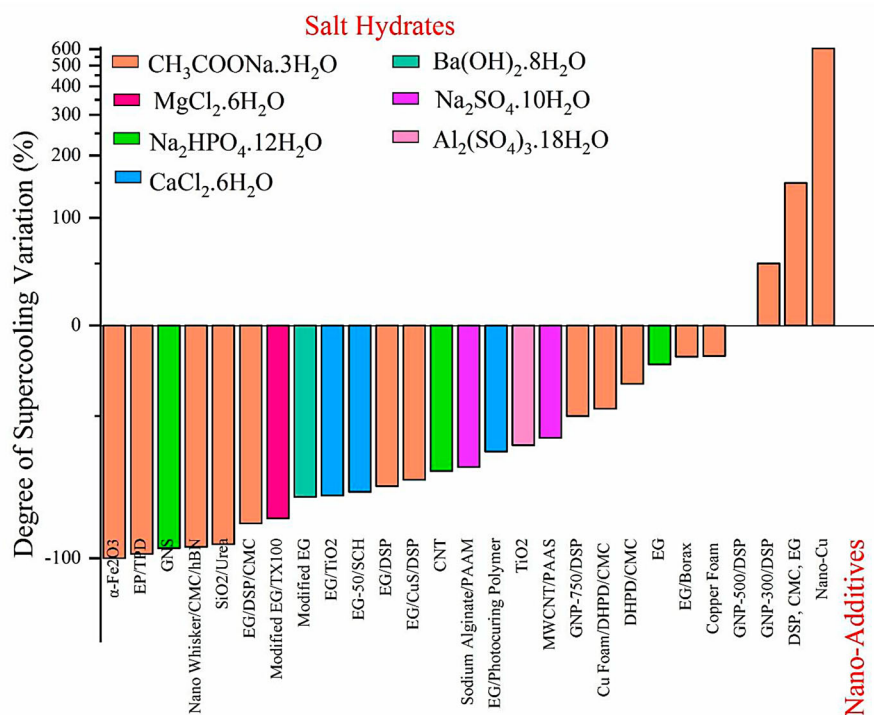


Figure 18. Variation in the degree of supercooling of salt hydrate due to loading of nano additives.

agent with salt hydrates avoids further phase segregation. On the contrary supporting materials incorporated to improve the thermal properties of salt hydrate also contribute towards mitigation of phase separation, due to the excess surface of supporting materials nucleating sites are developed. Supporting material acts as micro container resembling core-shell structure and hydrogels or resins and prevents salt hydrate solution from severe phase separation. Micro containers ensure the needy amount of water molecules around salt to improve uniform mixing. Mainly studied salt hydrate PCM is calcium chloride hexahydrate. With a reiterate number of thermal cycles, $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ experiences constant phase separation and forms $\text{CaCl}_2 \cdot 4\text{H}_2\text{O}$. Supply of excess water to initiate the formation of the hydrate by increasing solubility and by addition of nucleating agents, phase separation issues in $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ salt hydrate are diminished until 1000 thermal cycles.

Thermal cycling

Mechanism and factors influencing thermal cycling

In PCM, one thermal cycle refers to absorbing and releasing heat energy by salt hydrate once. Similarly to number of charging and discharging cycles in electrical batteries, number of heat charging and heat discharging cycles are important parameters for PCM. Thermal cycle reliability indicates the performance of inorganic salt hydrate after many thermal cycles. Similar to physical stability, thermal stability of salt

hydrates are crucial parameter for better utilization of TES. By measuring the thermal and physical properties of NDPCM before and after many numbers of thermal cycles, the thermal stability can be determined. In order to ensure the economic feasibility of salt hydrate PCM, thermal cycles must be as high as possible without deterioration in the property. Factors influencing thermal cycling are (a) absorption of moisture, (b) improper crystallization, (c) impurities and defects.

Effect on thermal cycling with composite additive

On application of pure salt as PCM for TES, there is a sharp decrease in heat storage potential of salt hydrate with an increase in a number of thermal cycles. Drop-in heat storage potential are predominant due to the following reasons (a) evaporation of water molecules (b) exudation of fluid and (c) phase segregation. However, the above issues are controlled with salt hydrate mixture due to close space structure, as they prevent the direct interaction of hydrated salt with the external environment and avoid overflow of fluid and evaporation of fluid. Thermal cycling consistent with salt hydrate can be improved by reduction of phase separation as both are directly related. Phase separation of salt hydrate is prevented by adopting the microencapsulation and confinement technique. Based on the stabilizing mechanism, followed by various encapsulation techniques, the thermal reliability also changes. Core-shell encapsulation technique of encapsulation is slightly effective than the incorporation of porous supports and 3D network structure, as core-

shell technique completely packs the salt hydrate. Encapsulation of $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ [142] using sol-gel slightly reduced the latent heat, in-situ mini emulsion technique of encapsulation adopted for $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ salt hydrate [7] ensured better thermal reliability after 100 cycles. An elaborated study on thermal cycling of $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ was performed by 1000 number of cycles [135]. The salt hydrate was studied without the inclusion of nano additives; results ensured variation in latent heat of salt hydrate by 10.4% for 10 thermal cycles, 6.4% for 100 thermal cycles and the attractive part was drop in 0.3% latent heat after 1000 thermal cycles. The results ensure that there is no particular trend or regular change in a thermal property of salt hydrate with an extended number of thermal cycles, however, reasons justifying the variation in latent heat of fusion were improper crystallization of salt hydrate as the testing quantity were taken in milligrams.

Application of composite additives for TES

With an intense to meet the future energy needs on a global scale, PCM are integrated with thermal systems to enhance TES. Improved TES tends to increase the potential of sustainable energy sources. Thermal energy changes in salt hydrate PCM during the transition of phase provide the potential to integrate salt hydrate PCM with heating and cooling of buildings. In addition to latent heat, the isothermal phase transition characteristic ensures its application in textile clothing's, BTMS and in electronic cooling applications. The thermal cycling ability of salt hydrate PCM offers TES facility in solar thermal applications like solar water heaters and solar air heaters.

Thermal conditioning of buildings

Energy demands for building heating and cooling, building construction are swelling every year due to the high usage of energy in developing nations. On an average, about one-third of the energy generated is consumed in the energy sector, and they contribute to 40% of CO_2 emissions globally. In order to improve the effective utilization of energy, PCM-based thermal storage systems are employed. Abundant salt hydrates such as chlorides and sulphates [169] are actively used in retaining the thermal performance of building heating and cooling. Application of salt hydrate in buildings for TES is analysed based on the development of materials, feasibility, thermal performance characteristic and net heat transfer capacity. Buildings are integrated with PCM in building sections like walls, floors, windows and roofs, they are also integrated using a portable module and PCM composite boards.

Wall

$\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ /expanded vermiculite shape stable PCM was combined with magnesium oxychloride cement (MOC), in three different methods such as multilayer board, pre-coated particles and direct mixing block on a lab scale [170]. Expanded vermiculite (EV) offers elevated heat storage potential to $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ salt hydrate in terms of heating and cooling. It extends the holding time in MOC with $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ /EV PCM model house by 93 min during heating process and 83 min during the cooling process compared to the ambient condition. In another research work, $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ - $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ was used as salt hydrates [66]. The salt hydrate PCM was impregnated into EV. The maximum capacity of EV to hold the eutectic salt was 60%. The samples were characterized using TGA and DCS to determine the thermal properties. The performance of the test room was evaluated by placing composite PCM in the form of wall boards. Due to the TES potential of the wall board, the maximum room reduced by 4.8°C . Owing to the impregnation of salt hydrate into EV, the chemical stability of composite increased and provided resistance to phase separation and leakage.

Floor

Salt hydrate PCM with PTT of $25\text{--}45^\circ\text{C}$ can be actively used for improving the thermal performance of building and to save energy costs. Heat fluctuation in buildings due to solar radiation can be reduced using salt hydrate PCM for TES. $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ are restricted in real-time applications due to the issue of supercooling and phase separation [94]. Flake graphite is an effective nucleating agent that reduces the supercooling effect; in addition, it also improves the thermal conductivity of the PCM. Along with flake graphite, absorbent polymers are employed to reduce phase separation in salt hydrate PCM. Phase separation and the supercooling issue of $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ salt hydrate PCM were suppressed by doping expanded graphite, the thermal characteristics analysis of nanocomposite PCM ensured suppression of supercooling and improved thermal conductivity [96]. Furthermore, the PCM were effectively used in floor heating applications by TES. Magnesium nitrate hexahydrate $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ with a latent heat value of $130\text{--}150\text{ KJ kg}^{-1}$ and low cost are extremely suitable for building heating applications [88]. A numerical computational simulation model was developed to analyse the heat transfer in building envelope considering $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ as salt hydrate PCM and graphene with a varying fraction of volume as nano additive [171].

Roof

Fu et al. [161] prepared a form-stable salt hydrate PCM with non-combustible characteristics to enhance

the safety measure while integrating with buildings. Salt hydrate of $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ (98%) and $\text{SrCl}_2 \cdot 6\text{H}_2\text{O}$ (02%) were adsorbed on layers of expanded perlite (EP). It was noted that perlite sheets effectively absorb 55% of salt hydrate to a maximum capacity. Thermal cycling analysis was performed for 1000 cycles to ensure the thermal stability of the composite. PCM boards were fabricated by placing a perlite sheet absorbed with salt hydrate into a bag followed by compression. Later the PCM boards were embedded within the brick and were used with the building roof to maintain the thermal stability of the buildings. The PCM-based roof provided better insulation as their thermal conductivity was lower than pure $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ (98%) and $\text{SrCl}_2 \cdot 6\text{H}_2\text{O}$ (02%) salt hydrate PCM. Building energy consumption varies with respect to season; hence to improve the performance of TES in buildings for better energy-saving potential, Ye et al. [172] experimentally studied the thermal performance of building using PCM. Salt hydrate $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ was initially mixed with 08% and 15% of $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ to form PCM1 and PCM2, respectively. Both eutectic salts obtained were combined with EG to form PCM panels and were experimentally studied by placing the panels in the roof and southern wall. Buildings with PCM panels showed better indoor temperature and reduction in the fluctuation of indoor conditions.

Portable module and PCM composite boards

For improving energy-saving and to reduce the temperature variation inside the buildings, Xie et al. [109] integrated salt hydrate PCM in the buildings. Leak-resistant eutectic salt composite $\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$ – $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ was prepared using two-step methods. PCMs were made into a panel with a layer of polyurethane acrylate followed by curing using UV rays. Experimental results showed that the comfort time inside the room with PCM panel increased from 3.85% to 92.10% compared to the room without PCM panel. Characterization of the salt hydrate diatomite PCM sample was carried out to ensure form, chemical and thermal stability using weight loss testing, Fourier transform infrared (FT-IR), TGA and DSC testing. PCM panels provide the flexibility of easy replacement, which is not possible with PCM incorporated with building cements and bricks. EP and EG have a prop up matrix, which makes it reliable for incorporating with PCM. $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ salt hydrates were embedded with EP [174] and EG [160] to form stable PCM boards, which could be placed inside the building to ensure thermal stability. Figure 19 shows the layout and testing profile of nano additive salt hydrate for buildings. The composite PCM boards were embedded into aluminium plates to ensure leakproof. Experimental testing was carried out by placing the PCM composite board

inside the room as well as in roofs of buildings. PCM boards also have the advantage of using as decorating roofs in buildings. In all the case thermal performance of buildings were drastically improved with incorporating of nanoparticle dispersed salt hydrate PCM. The low cost of the salt hydrates and non-flammable characteristics attract their application in numerous fields. The cost of organic PCM ranges from 1.88 to 53.9 US\$ kg^{-1} , whereas calcium chloride hexahydrate being the most commonly used salt hydrate cost only 0.20 US\$ kg^{-1} [175].

It is very clear that without a proper economic performance of inorganic salt hydrate for energy saving in buildings, investment decisions on the TES techniques would not be made. Thus the authors have reported few economic analyses carried out with inorganic salt hydrate PCM integrated buildings in real-time application. Bao et al. [94] carried out a numerical investigation on the performance of flake graphite/ $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ as TES for a building located in Hongkong and Changsha. From evaluation on payback period, it has been observed that the composite PCM of flake graphite/ $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ were 18.3 and 8.4 years for buildings in Hongkong and Changsha. The payback period is ensured to be well below the normal life span of residential buildings, which is 60 years. Tan et al. [176] presented an economic analysis using a mixed integer linear programming model for PCM storage system with SP11 salt hydrate for office building located in Sweden. 9380 kg of SP11 salt hydrate was used as inorganic salt hydrate PCM, cross-linked polymer sodium polyacrylate was added as a thickening agent to overcome the phase separation issue of inorganic salt hydrate PCM. As major industry sectors look for a short span of payback period. The cost of PCM for TES is estimated to be within 921EUR for effective economic benefit. Souayfane et al. [42] designed a TES wall using PCM by combining it with transparent insulation material and evaluated their energy and economic performance at different climatic conditions. Comparison between south-facing double glass transparent wall and PCM wall with transparent insulation material ensures the later to be cost effective and beneficial. For polar and subarctic regions, the payback period of PCM walls with transparent insulation material is calculated to be 10.5 and 7.8 years. Though the literature report shows economically beneficial results, these studies are limited to location-specific as the energy cost of natural gas and electricity varies from country to country, the economic situation of the nation's varies and each building are subjected to varying climatic condition. One important observation made is that only a handpicked number of research works on nanoparticles dispersed inorganic salt hydrate have carried out the economic analysis thus further research works are expected to develop industrially economically

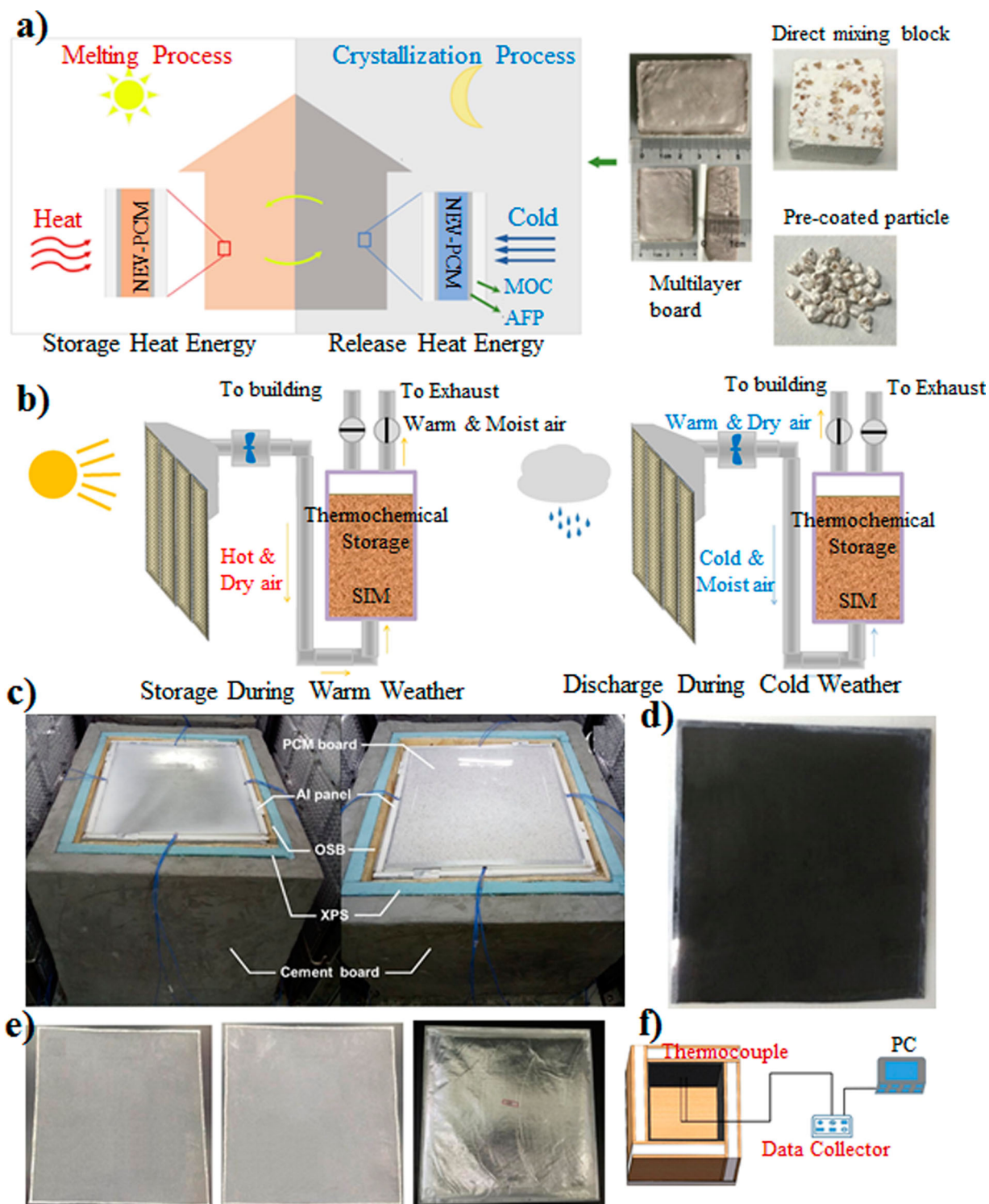


Figure 19. (a) Preparation and incorporation of NEV-PCM into buildings [170] (with permission from Elsevier), (b) charging and discharging of building thermal storage and heating [173] (with permission from Elsevier), (c) PCM board with $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ salt hydrate and EP [174] (with permission from Elsevier), (d) PCM board with $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ salt hydrate and EG [160] (with permission from Elsevier), (e) PCM board with $\text{CaCl}_2 \cdot 6\text{H}_2\text{O} - 08\text{wt}\% \text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}/\text{EG}$, $\text{CaCl}_2 \cdot 6\text{H}_2\text{O} - 15\text{wt}\% \text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}/\text{EG}$ [172] and (f) PCM panel incorporated testing unit with data logger (with permission from Elsevier).

feasible additive dispersed PCM with a short payback period.

Novel smart textiles

Garments are a vital part of all human beings, which are in direct contact with the body. Research hotspot

in the area of smart textile technology focuses on developing smart cloth materials for medical, military, aerospace and industrial applications that can provide warmth during winter and coldness during summer with the help of PCMs [177,178]. PCM are formed into microcapsules for its integration with fabrics. The commonly used technologies for incorporating

PCM into textile fabrics are (a) fibre technology, (b) coating, (c) lamination, (d) melt spinning, (e) injection moulding technique and (f) foam technique. The major drawback of incorporating salt hydrate with fibres is that the PCM can be washed away during wiping, abrasion and washing of cloths.

Incorporating salt hydrate PCM with fabrics causes three effects, namely, (a) heating effect, (b) cooling effect and (c) thermo regulating effect. Utilization of salt hydrate PCM in heat-protective gloves can avoid burn injuries to the worker [178]. PCM integrated glove will ensure the temperature of glove till the melting point of PCM is reached, as well it also extends the usage time of glove. Due to the non-combustible characteristic, salt hydrate PCM are encapsulated with silicon rubbers. Active thermal insulation effect are provided using PCM in skin wear clothing in addition to the passive effectives, thereby improving comfort. PCM also ensures moisture management and thermal comfort in chemical protective fabrics used in laboratories. Fire fighters are easily exposed to burn injuries as they face high heat flux during operation. The inclusion of PCM in firefighting garments increases the duration required to cause burn accidents [179]. The time taken by the PCM for its complete melting is the extended duration the firefighter gets before the occurrence of burn. The melting point of salt hydrate PCM are preferred based on the heat flux intensity like 50–150°C for low heat flux exposures and 50–200°C for medium and high heat flux exposure. The major discussion is on the optimal mass of salt hydrate PCM that has to be used in the garment to ensure minimal load of the garment for the comfort of fire fighters.

Transfer of heat and moisture in porous textile fabrics is a complex phenomenon. In addition to heat transfer by conduction, convection and radiation transfer of moisture through wicking, evaporation and diffusion also take place. Similarly, for PCM fabrics in addition to latent heat, the thermal performance of clothing is influenced by PCM quantity, method of incorporation and size of micro PCM. Thus, to analyse the overall thermal performance of PCM fabric, a number of indices like thermal regulation capability, thermal psycho sensory intensity (TPI) and static thermal insulation are characterized. For a PCM fabric, TPI expresses the thermal perception by the body and it can be measured using a fabric intelligent hand tester. TPI tends to increase with an increase in PCM level in the textiles [181]. Numerical analysis on garments with a hybrid cooling system integrated with PCM and ventilation fans were carried out by Wan et al. [180]. Figure 20 shows the designed garment with a hybrid cooling model and the different layers considered while numerical analysis and human body temperature with respect to PCM materials.

Garments with PCM exhibited lower body and core temperature compared to garments with ventilation fans, due to the wide surface spread area, however, the hybrid system were much effective than the individual cooling techniques. The garments also exhibited issues with absorb of heat from the environment and condensation of moisture on the PCM surface. In addition to the above application, PCM has also left footprint in (a) space suits for the thermal comfort of astronauts, (b) sportswear like cycling underwear, snowboard gloves and athletic wears, (c) bedding accessories like pillows and mattresses for thermal comfort, (d) polyethylene glycol treated cloths are used in medical applications, (e) shoes and accessories of militants to overcome cold climate and (f) in automobile seat covers [182].

Electronic gadgets

Major issues with electronic gadgets are heat generation and dissipation, as the high temperature of the devices directs to burnout of circuits, failure of integrated circuits and accidents leading to improvised safety systems [183]. However, research's on integrating desirable PCM with electronic equipment's has been the hotspot as they provide better thermal protection and they are a passive cooling technique. Alimohammadi et al. [184] carried out an experimental analysis to understand the performance of salt hydrate PCM $\text{Mn}(\text{NO}_3)_2$ by doping Fe_3O_4 as a nano additive. The nanoparticle was effective to decrease the electronic chip temperature to 10.5°C, whereas for pure PCM, it was 14°C. The presence of nano particle increases the thermal conductivity of the composite and increases the heat dissipation rate for effective cooling. The numerical study was carried out using salt hydrate S44 to analyse the thermal performance in a heat sink. In the model generated salt hydrate S44 composite was compared with the lauric acid nanoparticle, as S44 composite transferred high heat per fluid pumping power, it exhibited better thermal performance [185]. It is very keenly noted that due to the corrosive nature of salt hydrate and uncertainty toward long-term reliability, application of salt hydrate PCM for thermal heat dissipation of electronics are less preferred compared to the organic and metal PCM's. Glauber's salt is a common PCM used in heat sinks and micro-channels of electronic devices but are limited to devices where the number of thermal cycling is very less unless it's encapsulated. Further research works on integrating nano additives dispersed salt hydrate PCM are expected to study the effect of corrosion inhibitors with salt hydrate for reliable use of PCM with electronic gadgets. Phase separation has to be completely addressed for real-time application.

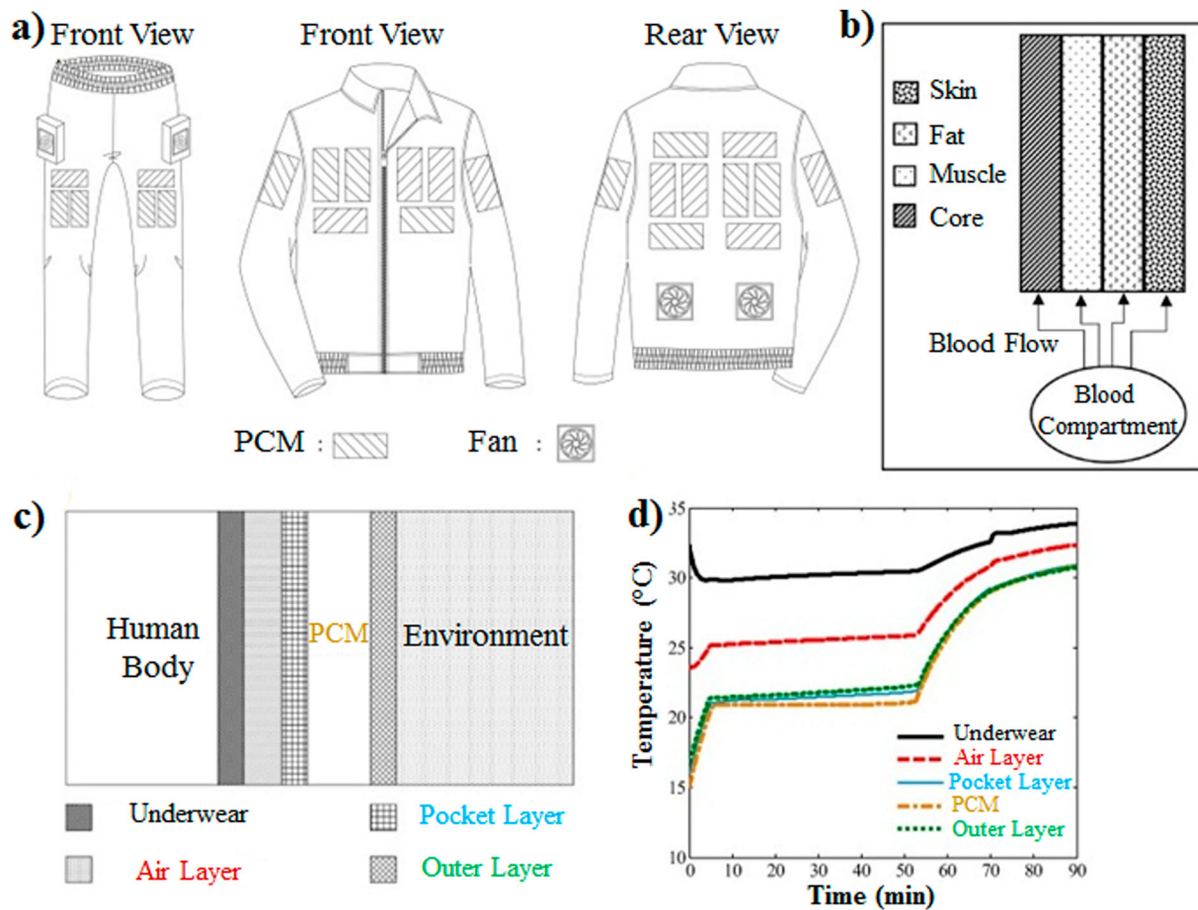


Figure 20. (a) Garment designed with PCM layer and ventilation fan, (b) human body barrier, (c) garment with PCM layer and (d) body temperature variation [180] (with permission from Elsevier).

Battery thermal management system

Temperature dependence on battery plays a vital role which decides their performance. Operating battery at lower temperature leads to a decline in several parameters like charge acceptance, power capacity, life span and round-trip efficiency [186,187]. On the other hand, operating battery at a higher temperature contributes to several issues like rapid self-discharge and capacity fade [188]. In order to achieve better performance battery should be operated at an intermediate temperature. Lithium-ion batteries are very sensitive to temperature, thus a suitable BTMS system needs to be implemented to ensure the optimum operating range of electrical battery [189]. Battery thermal management is carried out by air cooling system [190], direct and indirect liquid cooling system [191], thermo-electric heating-cooling [192] and PCM [193]. PCM integrated BTMS system due to its passive energy utilization to cool battery seems to be a smart option. PCM absorbs the heat generated during the charging and discharging process due to its phase changes phenomena, and it prevents further temperature rise naturally. Peak temperature loads of lithium-ion batteries are reduced by employing PCMs. Performance of organic PCM like paraffin and fatty acids like capric acid, lauric acid, dimethyl sabacate

and vinyl stearate are experimentally studied by various researchers compared to inorganic PCM as they offer outstanding thermal reliability. The low thermal conductivity nature of paraffin was modified by employing carbon fibre additives in different proportions. The experimental analysis also reflects that blending 2 mm long carbon fibre with 0.46 wt-% into PCM can effectively reduce the peak temperature rise by 45%. In addition, the corrosive nature, super-cooling characteristic and dehydration are some serious issues, which further makes inorganic PCM less competitive to organic PCM to be applied for thermal management. Thermal runaway is a phenomenon during which an exothermal process prompts the other processes, which subsequently result in an unmanageable rise in temperature. Thermal runaway commonly occurs in batteries and leads to fire in several cases.

Cao et al. [194] employed expanded graphite doped with SAT to prevent the thermal runaway issue faced by lithium-ion batteries. Due to the potential of SAT to store heat in two stages like (a) latent heat storage up to a temperature of 58°C and (b) thermochemical heat storage up to a temperature of 106–140°C, they predominantly reduce the overheating of the battery and maintains the temperature under control. Lithium-ion BTMS was integrated with EG and

organosilicon blended inorganic SAT as PCM [195]. EG improved the thermal conductivity of PCM and organosilicon ensures proper leakproof for long-term application. As inorganic PCM are highly non-flammable, they provide high safety to battery systems than the organic PCM. Ling et al. [196] presented an innovative method of heat up lithium-ion battery using $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ salt hydrate with negligible energy utilization. Heat storage and release from the salt hydrate PCM is initiated using a specially designed triggering device which is based on Gibbs function. The local pressure of PCM was used developed using a stick and spring mechanism in the triggering device. CMC is used as a thickening agent to analyse the stability of supercooling. The effect of supercooling always causes a negative impact on the application of salt hydrates, whereas this particular work has used supercooling of PCM for controlling heat storage and release. Experimental result shows the heating of lithium-ion battery at a rate of $7.5^\circ\text{C min}^{-1}$. Composite PCM with expanded graphite impregnated over the surface of salt hydrate with a mixture of $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ and $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ at 41.3:58.7 wt-% were used to perform an elaborative study on electrical cycling and thermal runaway penetration of lithium-ion batteries [197]. Salt hydrates PCM were filled into the battery pack for enhanced thermal management. A comparative analysis of the system was carried out using organic and inorganic salt mixture shows that inorganic salt hydrates successfully restrict thermal runaway propagation in lithium-ion batteries. Ramandi et al. [198] carried out a numerical study to understand the thermal performance of $\text{Na}_2(\text{SO}_4) \cdot 10\text{H}_2\text{O}$ and $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$. The analysis model shows effective performance of the battery when double-shell PCM insulations are adopted. Hybridizing PCM salt hydrated (HM-S29), heat pipe and a spray loop, a cooling system for thermal management of LiFePO_4 battery was carried out by Lie et al. [199]. During day time operation of the battery, the heat generated was actively absorbed by salt hydrate for effective functioning. Figure 21 shows the application of inorganic salt hydrate PCM, SAT and $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ along with their additives for proper thermal management of lithium-ion batteries. Safety test evaluations of SAT in the presence of fire shown in Figure 21(d) ensure its fire resistance property. Due to the non-explosive and non-flammable nature of salt hydrate, they find themselves to be tested and used in electric vehicles. Organic PCM like paraffin's are most commonly used in BTMS owing to high heat capacity and availability of organic PCM materials with a different melting point. Though inorganic salt hydrate PCM exhibits enhanced safety properties and advanced thermal properties than organic PCM, the supercooling and phase separation nature of salt hydrate makes it unpopular among researchers for

BTMS application. However, nanoparticle doped salt hydrate with reduced supercooling effects are numerically tested.

Solar thermal system

Owing to the fact of inexhaustible and eco-friendly nature of solar power, they are considered to be the most abundantly available renewable energy; however, challenges due to climate, weather and cyclic day and night, uninterrupted functioning of solar appliances are a research hotspot. Solar thermal systems integrated with PCM provide a facility to store thermal energy. Thermal energy stored using salt hydrate PCM during the day time in the presence of sunlight can be utilized during late night or in the absence of solar power without further dependence on electrical energy produced by conventional fossil fuels for heating application [200]. Extended usage of the solar thermal system during the intermittent period of solar energy by salt hydrate PCM reduces the usage of fossil fuels which, on the contrary, reduces environmental pollution [5]. Application of PCM with the solar thermal system is considered to be a reliable technique to provide the continuous power supply. PCM are very attractively incorporated with a solar thermal system of all operating temperature ranges. The selection of PCM depends on its melting point and latent heat. Micro and nano additive PCM are becoming a significant part in solar thermal appliances. These include the application of salt hydrate with solar domestic and solar industrial systems, which is discussed in this section.

Varol et al. [201] incorporated $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ as PCM on solar flat plate collectors and performed both experimental and numerical investigation on its performance. Results ensured enhanced performance of solar water heated with salt hydrate PCM compared to the conventional flat plate collector. Using $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ and $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ as PCM experimental comparative analyses on the performance of solar concentric parabolic cooker were studied [202]. A new design was made with the cooker receiver with 25 PCM tubes to fill salt hydrate. Novel research focused on solar thermal power generation with thermal energy storage for the temperature range of 120–300°C was carried out by Tamme et al. [203]. Alkali metal nitrate and its mixtures are tested for its thermal performance with the expanded graphite in three different design enclosures (encapsulated metal tubes, expanded graphite heat exchanger tubes, graphite foiled heat exchanger tubes). Using eutectic salt hydrate mixture of NaNO_3 and KNO_3 in the annular space of the receiver tube of solar parabolic trough collector (PTC), experimental analyses were carried out [204]. The salt hydrate PCM provide supplementary thermal energy from the PCM materials at the absence

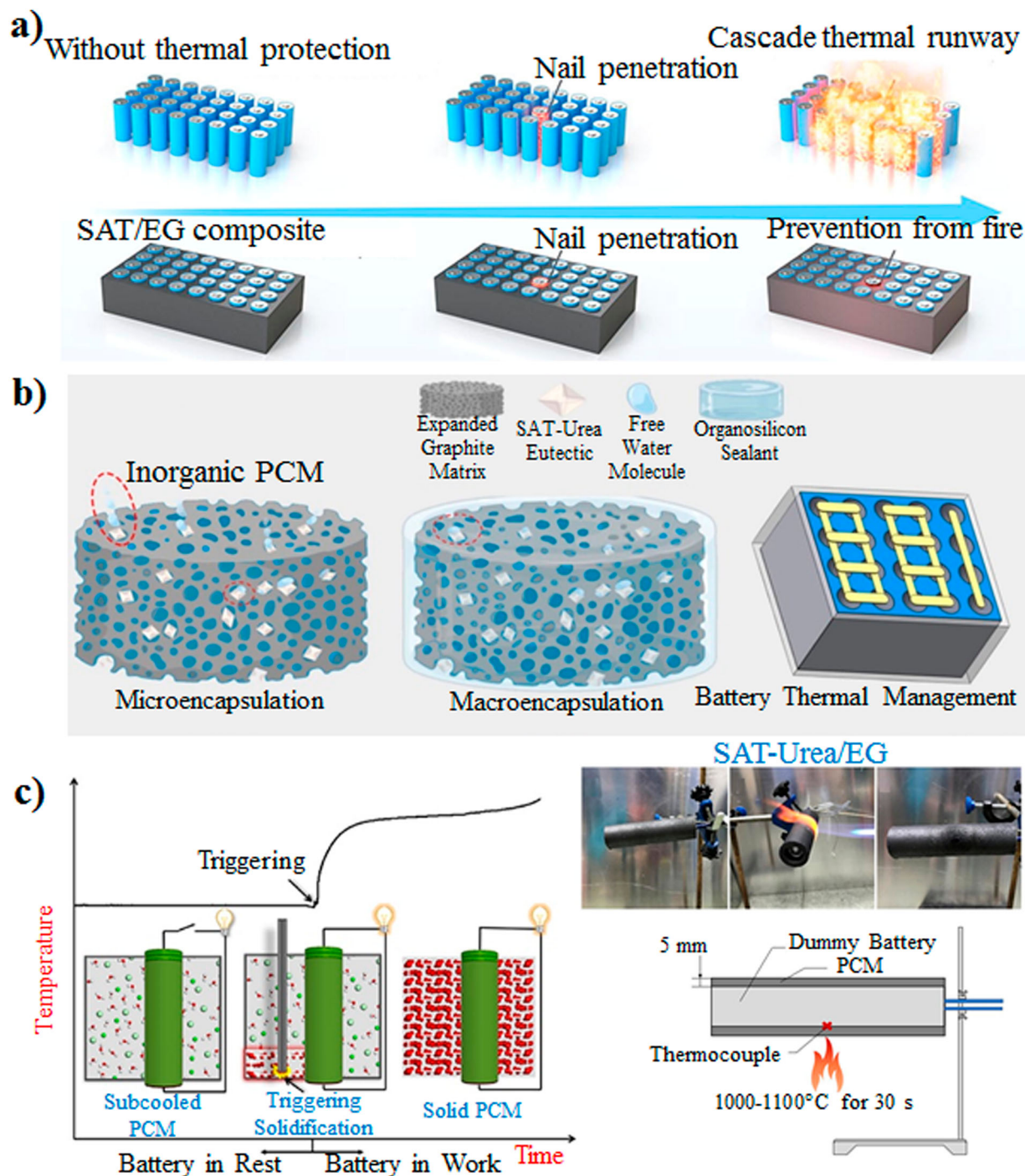


Figure 21. (a) Thermal runaway prevention of lithium-ion battery using SAT [194] (with permission from Elsevier), (b) SAT-urea EG PCM composite for BTMS [195] (with permission from Elsevier), (c) passive heating strategy of lithium-ion battery using inorganic PCM with a trigger device [196] (with permission from Elsevier) and (d) performance evaluation of SAT-urea PCM when exposure to accident fire [195] (with permission from Elsevier).

of solar radiation. It was observed that the temperature and working fluid and thermal efficiency of solar PTC improved with integrating thermal batteries. From numerous literature searches, it is very clear that integration of the solar thermal system with organic PCM has covered about 90% of the solar domestic systems and eutectic PCM have covered 80% of the industrial system like solar parabolic and solar dish power plants. Solar thermal system requires salt hydrate that is high stable for a minimum of 1000 number of thermal cycle for better operation, hence research on

nano additive doped salt hydrate with stability for high number of thermal cycle are highly impactful.

Though the use of solar thermal system enhances environmental protection, the use of PCM with the solar thermal system is not often economically feasible, especially when it comes to the integration of nanoparticle dispersed PCM. The above issues are due to the high price of PCM and nanoparticles. The higher cost of PCM will continue to exist until large-scale markets are developed for the preparation of PCM and for hybridization of nanoparticle with

PCM on a large scale. In addition to the above, government subsidy could be another effective way to reduce the cost of PCM and its additives for improving TES. Plenty of research work on PCM integrated solar thermal system are carried out, never the less there is still a lack of research work discussing on the economic analysis of nanoparticle dispersed inorganic salt hydrate for the solar thermal system.

In this section, the authors have discussed on the advanced application of nano additive-enhanced salt hydrate PCMs in the field of building heating and cooling, novel textile materials, electronic gadgets, battery thermal management and solar thermal systems. From the above analysis, it has been observed that for thermal management of buildings, the quantity of salt hydrate required is of high due to large volume of space cooling and thus, any salt hydrate PCM with higher latent heat, within melting point temperature range of 28–35°C are highly preferable. Low-cost nano additives that provide better thermal stability and lower degree of super cooling are highly recommended. On an average, the number of thermal cycles that would take place in building units is expected to be 2 cycles per day, hence any salt hydrate with better thermal stability for 1800 cycles is expected to provide better performance for almost 3 years. Whereas salt hydrate PCMs in textile materials need high resistance to solubility, and they are expected not to react with sweat. The need for thermal stability for the nano enhanced salt hydrate PCMs can be rather lower since people use numerous cloths, and the same cloth is expected to be used repeatedly only in rare cases. For electronic application, salt hydrate must not cause corrosion as they would be subjected to direct contact with metal products, the thermal conductivity of the composite PCM used for electronic gadgets are expected to higher, to ensure better cooling of the gadgets. Similarly, BTMS units are one, which are widely explored in recent days, salt hydrates are yet to be explored as for major BTMS units, paraffin and organic PCM are highly preferred due to their well-developed technique. The last application discussed is the solar thermal systems, which make use of medium and higher temperature salt hydrate PCM, as they contribute towards thermal energy storage. Higher the thermal stability nature of nano composite salt hydrate PCM, higher is the application with solar thermal systems.

Conclusions and future challenges

Conclusions

Nano additive-based salt hydrates can effectively prevail over the drawback of thermal conductivity, supercooling and phase separation of conventional PCMs, which makes them highly capable for TES

applications. The significance of salt hydrates has attracted numerous researches in the recent decade. In this review article, the authors specifically summarized the advances in nano doped salt hydrate, thermal performance and their smart applications. Following are the conclusions and authors' comments on nano additives dispersed salt hydrate PCMs;

- Regarding doping of nano additive into salt hydrate, mainly three materials are considered (a) metal additive which exists in the form of powder and metal foams; (b) carbon additive which is classified based on their dimension as 0D, 1D, 2D and 3D under which fullerene, CNT/CNF, graphene and graphite nano additives are used; (c) polymer additives. Metal nano composite in the form of metal foam reveals a dominant role in improving the heat conduction rate as well it weakens the super cooling effects of salt hydrate. Graphene, due to its higher thermal conductivity and surface area, makes a notable impact on doping with PCM. Carbon additives are the major nano additives that are doped with most of the salt hydrate PCMs achieving enhanced thermal properties. However, the lack of studies on the hydrophilia nature of multi-dimensional carbon additives with salt hydrate PCMs are not scientifically discussed by various authors.
- Numerous hybridization techniques exist for the inclusion of metal, polymer and carbon-based nano additives into salt hydrate; however, the highly adoptable technique for dispersion of additive into salt hydrate PCMs are; (a) melt and direct mixing; (b) vacuum impregnation method; (c) direct mixing followed by ultrasonication and magnetic stirring. For preparing solid pills, the sintering process is mostly followed by researchers. All the research works carried out so far in regard to characterization and enhancement of thermophysical properties are only for lab-scale application, and there is a variation in supercooling value as reported by different authors with respect to different binding agents and additives.
- Very limited research on the influence of morphological structure, size of nano additive, the concentration of nano additive on the thermophysical characteristic of salt hydrate have been performed.
- Thermal conductivity of salt hydrate PCMs was observed to increase by almost 2000% on the dispersion of EG, however, research work by certain authors shows a decrease in thermal conductivity of salt hydrate PCMs with a dispersion of EP. Similarly, latent heat of salt hydrate PCMs dropped by a maximum of 54%, however, majority of nano additives reduced the enthalpy change by 10–20% which was under permissible value. Variation in melting point of salt hydrates were between –20% and +20%. Complete suppression of supercooling

effect was observed with the inclusion of α -Fe₂O₃ with SAT, but there is a lack of work with the same metal additive doped with other salt hydrates to study the supercooling effect. But supercooling effect increased by 600% on the dispersion of nano-Cu with salt hydrate PCM as they are non-nucleating agents. Most of the studies highlighting the reduction of supercooling in salt hydrate with nucleating agents have limited their testing with less than 50 numbers of thermal cycles, ensuring a lag in the reliability of nucleating agents for long-term application. In addition, the crystal structures are yet to explore in detail.

- An application point of view, nano-salt hydrate PCM is widely used in building heating and cooling application due to the low cost of salt hydrate. Owing to the corrosive nature of salt hydrate, they are yet to be explored with electronic gadgets. Every research work is initiated to transfer the demerits in every theory and application into merits. One such drawback of salt hydrate PCM is the degree of supercooling. Heat storage and release of PCMs are always temperature-dependent. On the contrary, utilization of supercooling effect of salt hydrate PCMs for enhanced thermal management of electrical battery are an outstanding research work which operated by a trigger mechanism to initiate the heat release and storage as per the need.

Future challenges

Despite several advanced research progress in the synthesis and development of nano additive salt hydrate PCMs, quite a few aspects of future research are highly recommended to pursue the betterment of the research community as given below;

- Current researches are mainly focused on integrating nano additive with salt hydrate PCM in a laboratory scale and performing characterization testing of their thermal property. However, nano additives PCM are used in building materials for thermal comfort of human beings. In addition, many real-time experimental application-oriented research works are to be carried out to propel the commercialization of salt hydrate PCMs.
- On the emerging hybridization techniques, nano additives are doped with salt hydrate using techniques like impregnation, direct mixing and ultrasonication. But no hybridization techniques or methods have been reported yet on doping of nano additives with PCM for large-scale application. The major problems associated with such hybridization techniques are (a) non-uniform distribution of additives over PCM; (b) possibility of

hydrate to get evaporated at atmospheric condition; (c) highly feasible for corrosion reaction due to the atmospheric oxygen content. Considering the above-mentioned factors lab-scale hybridization techniques need to be explored. As per the analysis and literature review, the authors suggest that a direct mixing technique with a stirring process is preferable for large-scale applications. Alongside with fast-growing graphene industries worldwide, salt hydrate PCM with graphene additive is expected to make a remarkable contribution.

- Though GO nano sheets make a remarkable contribution towards enhancing the thermal property of salt hydrate PCM. GO nano sheets have poor dispersion ability and get easily agglomerated. To overcome agglomeration of nanoparticles, additives like surfactants are added. However, this technique may change the base property of salt hydrate NDPCM and may affect the performance. Thus, there is a need for external mechanism to avoid agglomeration and the poor hydrophilia nature of carbon additives.
- Dispersion of CNT, SWCNT and MWCNT with salt hydrates are an important aspect for the enhancement of thermal properties of PCM. The hydrophobic characteristic of carbon material makes them less dispersion in salt hydrate and gets entangled with other carbon particles. Thus CNT, SWCNT and MWCNT additives are made hydrophilic by functionalization using macromolecule; however, the studies carried out so far are only with organic PCMs, whereas inorganic PCMs are yet to be discussed.
- Higher manufacturing costs of nano additives are notable issues that hinder the commercial application of salt hydrate. Economic studies considering the thermal performance, environmental benefits over conventional systems and overall cost of the system with net pay back period would be an insight for attracting high investing stakeholders. So far, no special article provides detailed economic benefits with proper accuracy considering all the operational, economic and environmental parameters.
- Lack of theoretical modelling on nano structured PCM to yield empirical relations to predict the thermal properties of NDPCM are limited for the combination of only certain nano additives and inorganic salt hydrates. Literature also lags theoretical mechanisms for understanding the enhancement and deterioration of properties like supercooling, thermal conductivity, latent heat and viscosity on doping of nano additives.
- Salt hydrate PCMs are applied for long-term applications that are justifiable with thermal cycling tests. However, recent studies on thermal cycling performance of nano additive salt hydrate are

very minimal. Thermal cycling analyses are performed for only few hundreds of cycles to ensure thermal and chemical stability which is very far away from real-time applications. Better thermal stability with a low degree of supercooling and less leakage are required for more exploration.

- Similar to the trigger mechanism adopted in lithium-ion battery thermal management, a desirable heating control strategy of energy storage and release from salt hydrate PCM can be applied to building application, solar thermal cooking application, smart textiles and electronic gadgets based on Gibbs function, which is discussed.
- Existing advanced research work ensures confinement of organic PCM for offering enhanced stability over leakage issues and low thermal conductivity; on the other hand, this technique is not adopted for inorganic salt hydrate PCM due to evaporation of water from the pores during phase transition. The evaporation of water affects the reversibility of thermal cycling.

Disclosure statement

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